MIL-C-52640A(ME)
20 June 1973

SUPERSEDING
MIL-C-52640(ME)
5 March 1969

MILITARY SPECIFICATION

CRANE-SHOVEL, CRAWLER MOUNTED, 12-1/2-TON AT 12-FOOT RADIUS, 3/4 CUBIC YARD, DIESEL-ENGINE-DRIVEN, FOR MILITARY OPERATIONS

1. SCOPE

1.1 This specification covers a diesel-powered crawler-mounted crane-shovel suitable for use as a crane, clamshell, dragline, shovel, backhoe, wrecking ball or piledriver. The crane-shovel shall be one size only with a rated lifting capacity of 12-1/2-tons at a 12-foot-load radius.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
RR-W-410	- Wire Rope and Strand.
VV-F-800	- Fuel Oil, Diesel.

FSC 3810

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ZZ-H-428	- Hose, Rubber: Preformed and Straight; Hose, Preformed: Flexible, Wire-Reinforced (for the Coolant Systems of Automotive and Other Liquid-Cooled Engines).
GGG-B-485	- Block, Tackle, Wire Rope, with Swivel Hook; (for Crane Use).

Military

MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for the Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-H-2514	- Hammer, Piledriver, Drop; 1,800 and 3,000 Pounds.
MIL-H-3199	- Heater, Vehicular Compartment 28 Volt DC.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and Their Attachments, Packaging of.
MIL-L-3661	- Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.

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MIL-S-3785	- Starter, Engine, Electrical: 24-Volt Direct Current.
MIL-G-3787	- Glass Laminated, Flat; (Except Aircraft).
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-L-6085	- Lubricating Oil: Instrument, Aircraft, Low Volatility.
MIL-S-8660	- Silicone Compound.
MIL-H-8796	- Hose, Airduct, Flexible, Aircraft.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-L-11025	- Lead, Pile-Driver; with Catwalk.
MIL-B-11040	- Belt, V; Engine Accessory Drive.
MIL-E-11276	- Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-H-13071	- Housing, Lamp, Resilient Mounting, for Construction Machinery.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty, Single-Conductor and Multiconductor.
MIL-C-13516	- Circuit Breakers; Manual and Automatic (28-Volt).
MIL-H-13531	- Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible).
MIL-H-13910	- Hydraulic Fluid, Polar Type Automotive Brake, All-Weather.
MIL-C-14055	- Cup, Hydraulic Brake Actuating Cylinder: Synthetic Rubber (General Specification for).

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MIL-R-14328 - Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).

MIL-I-15126 - Insulation Tape, Electrical Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.

MIL-B-16028 - Buckets, Clamshell.

MIL-B-16029 - Buckets, Dragline.

MIL-H-21103 - Hammers, Piledriver, Self-Powered, Diesel.

MIL-B-23276 - Ball, Wrecking.

MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.

MIL-S-40626 - Sign Kit, Vehicle Class.

MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

MIL-T-46786 - Tanks, Fuel, Engine, General Requirements for.

MIL-T-52176 - Teeth, Surface Ripping; (Points, Shanks, Keepers and Keeper Locks), Bucket and Dipper, Crane-Shovel.

MIL-T-52198 - Taglines, Crane and Crane-Shovel, Spring-Loaded Reel Type.

MIL-B-52203 - Boom, Crane: 20-Ton, Rough-Terrain; 20-Ton, Truck-Mounted; 12-1/2-Ton, Crawler-Mounted.

MIL-B-52207 - Boom, Jib, Crane, 15-Foot-Long, for 3/4-Cubic-Yard (12-1/2-Ton Cranker Mounted, 20-Ton Wheel Mounted) Cranes.

MIL-A-52363 - Air Cleaner, Intake: Dry-Type (for Internal-Combustion Engine).

MIL-E-52649 - Engine Cold Starting Aids, Ether Fuel-Primers.

STANDARDS

Federal

FED. STD. No. 595 - Colors.

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STANDARDS

Military

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-421	- Chain, Roller; Power Transmission and conveyor, Flat Link Plates; Base Pitch, Single and Multiple Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Sidebars, Single-Pitch.
MIL-STD-425	- Chain, Silent; 60 Degree Inverted Tooth, Power Transmission.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-646	- Electrical Circuit (Wire Marking) Identification for Tactical Military Vehicles.
MIL-STD-701	- Lists of Standard Semiconductor Devices.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS15795	- Washer, Flat-Metal, Round, General Purpose (In./MM).
MS17165	- Lamp, Incandescent, PAR-46 24 Volt Floodlamp, Mogul End Prong Base.
MS21322	- Clamp, Loop - Aluminum, Plain and Cushioned.
MS21333	- Clamp, Loop - Steel, Plain and Cushioned.

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MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS24532	- Indicator, Voltage; Battery-Generator 28 Volt DC.
MS24537	- Transmitter, Temperature Electrical Resistance - 24 Volts DC.
MS24538	- Transmitter, Pressure - 24 Volt DC 60 PSI.
MS24539	- Transmitter, Pressure - 24 Volt DC 120 PSI.
MS24540	- Indicator, Pressure 24 Volt DC, Range 0 - 120 PSI.
MS24541	- Indicator, Pressure - 24 Volt DC, Range 0 - 60 PSI.
MS24542	- Indicator, Temperature, Electrical Resistance - 24 Volt DC, Range 160° - 320° F.
MS24543	- Indicator, Temperature, Electrical Resistance - 24 Volt DC, Range 120° - 240° F.
MS24544	- Indicator, Liquid Quantity - 24 Volt DC, Engine Fuel.
MS27142	- Connector, Plug, Electrical-Pin Contact No. 12, 14, & 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical; Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical; Socket Contact, No. 14 & 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical; No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way No. 12, 14 & 16 A.W.G. Waterproof.

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MS27148	- Contact; Pin, Electrical Connector; No. 12, 14 & 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical-Pin Contact, No. 12, 14, & 16 A.W.G., Waterproof.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35340	- Washer, Lock-Spring, Helical, Extra Duty (Extra Heavy) Series.
MS35425	- Nut, Plain, Wing, UNC-2B.
MS35647	- Padlocks, Key Operated.
MS35829	- Hinge, Continuous (Piano), 0.075 Thickness, 0.187 and 0.250 Pin Dia.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS35916	- Tachometer, Mechanical, Fixed Mounting (4000 RPM).
MS39020	- Band, Marker, Blank.
MS39060	- Switch, Rotary, 28 Volt DC, Waterproof.
MS39061	- Switch, Toggle - SPST, SPDT, 24 Volt, DC, 25 Amp (Waterproof).
MS51074	- Horn, Electrical - Automotive Type, 24 Volt DC.
MS51967	- Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated, UNC-2B (In./MM).
MS51968	- Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNF-2B (In./MM).
MS53046	- Retainer, Battery - Automotive Tactical Vehicles.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS75057	- Connector, Plug, Electrical-Slave, 2 Contact, 24 Volt, Waterproof.
MS75058	- Connector, Receptacle, Electrical-Slave, 2 Contact, 24 Volt, Waterproof.
MS90530	- Switch, Pressure-Warning, SPST, Waterproof, 24 Volt, DC.

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MS90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.
MS90726	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A.
MS90727	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, UNF-2A (In./MM).
MS90728	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, UNC-2A (In./MM).

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D1384 - Glassware Corrosion Test for Engine Antifreeze
Quality Hot-Rolled Carbon Steel Bars

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Bruen Street, Chicago, IL 60605.)

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AMERICAN WELDING SOCIETY

Standard Qualification Procedure AWS-B3.0.
Welding Handbook.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL BRUEAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

SOCIETY FOR NON-DESTRUCTIVE TESTING

Handbook Vol. II.

(Application for copies should be addressed to the Society for Non-Destructive Testing, 914 Chicago Avenue, Evanston, IL 60202.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

ANS Z26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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REINHOLD PUBLISHING CORPORATION

Engineering Alloys.

(Application for copies should be addressed to Reinhold Publishing Corporation, New York, NY 10022.)

DEPARTMENT OF TRANSPORTATION

DOT Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents , U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The crane-shovel shall consist of a crawler-mounted, full revolving superstructure and front-end attachments. Each crane-shovel shall be complete and ready for operation. Each crane-shovel shall be capable of being converted for use in crane, clamshell, dragline, piledriving, wrecking ball, shovel and backhoe operations. The crane-shovel shall be constructed so that conversion to any one of these uses can be effected in the field without the use of special tools or the necessity of installing additional shafts, drums, clutches, or other parts that should for reasons of practicability be included at time of manufacture. The crane-shovel, equipped with boom, shovel front, or backhoe attachment, shall be capable of traveling over rough ground or under adverse or severe conditions. All operational functions for a given attachment shall operate independently and simultaneously where required for maximum performance.

3.1.1 Weights and dimensions. The physical characteristics of the crane-shovel without boom shall be as specified in table I.

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Table I. Crane Dimensions

Overall height, maximum	130 inches
Overall width (with catwalk removed), maximum	124 inches
Overall length, maximum	192 inches
Ground clearance, minimum	9-1/2 inches

The gross weight of the crane-shovel in operable condition and fully serviced shall be not less than 44,000 pounds and shall not exceed 54,000 pounds.

3.1.3 Compatibility. The crane-shovel shall be compatible with front-end attachments as specified in 3.9.1 and 3.15.

3.2 First articles.

3.2.1 Preproduction model. The supplier shall furnish one or more cranes for examination and testing within the time frame specified, to prove prior to starting production, that his production methods and choice of design detail will produce crane shovels that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.2.1.1 Engine. The crane engine shall be tested as specified herein (see 4.6.2.2), prior to being installed in the preproduction model and shall be subject to surveillance and approval by the Government.

3.2.2 Winterized preproduction model. When winterized crane-shovels are specified (see 3.16) in addition to nonwinterized crane-shovels, two preproduction models shall be furnished for examination and test. One of these preproduction models shall be a nonwinterized crane-shovel and one shall be a winterized crane-shovel.

3.2.3 Material identification. At the time the preproduction model(s) is furnished for examination, the supplier shall furnish to the Government, in the form of sketches or outline drawings (dimensions not required) identification of high strength constructional steels (steels requiring special process controls as defined in MIL-STD-1261) used in the

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fabrication of the crane-shovel and the recommended welding repair procedures for accomplishing repair of such steels by the military (see 6.4). Identification shall include type of steel with ASTM designation and grade and location by part or assembly.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more cranes for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Unless otherwise specified herein, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the crane-shovel by the Government.

3.4.1 Common parts. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts.

3.4.2 Hose. Hose and hose assemblies shall conform to the following:

(a) Coolant hose shall conform to SAE J20, heavy-wall, heavy-duty type (SAE J20R1). Sizes of hose shall be as recommended by the engine and radiator manufacturers. Hose clamps shall conform to SAE J536, type F, style 1, 2, or 4.

(b) All other hose shall conform to MIL-H-13531, and the fittings for this hose shall conform to the following:

Hose diameter	Fitting
1/4 in. through 3/4 in.	SAE J516 female 37-degree flared.
1 in. through 2 in.	SAE J513 with 4-bolt clamp on hose ends.

3.4.3 Breathers. All breathers for lubricant and fluid sumps on the crane-shovel shall preclude the entrance of water and air-borne dust. Breathers on fuel tank shall terminate at a level above the fill cap and shall be external to the superstructure.

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3.4.4 V-belts. V-belts shall conform to MIL-B-11040, and all belts for multiple belt drives shall be furnished as matched sets in accordance with MIL-B-11040.

3.4.5 Heater ducts. Heater ducts for air shall conform to MIL-H-8796.

3.4.6 Electrical cable. Low voltage electrical cable not covered by component specifications shall conform to MIL-C-13486.

3.4.7 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-T-704.

3.4.8 Threaded parts. All capscrews 1/4 inch nominal size and larger shall be in accordance with MS90725, MS90726, MS90727, or MS90728. All nuts 1/4 inch nominal size and larger shall be in accordance with MS51967 or MS51968. Lockwashers shall be in accordance with MS35340. Grade B nuts shall be used with grade 5 capscrews and grade C nuts shall be used with grade 8 capscrews. All other threaded parts shall be in accordance with Handbook H28 except when otherwise specified in specifications referenced herein. No self-tapping threaded parts shall be used.

3.4.9 Hydraulic cups. Synthetic rubber hydraulic brake cylinder cups for master and slave cylinders shall conform to MIL-C-14055.

3.5 Safety, human factors and sound level.

3.5.1 Safety. All exposed parts which are subject to high operating temperatures shall be insulated, fully enclosed, or guarded. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel and subject to breakage and entanglement of wire ropes shall be fully enclosed or guarded. Provisions shall be made to minimize dripping of grease or oil onto the superstructure floor. Protective devices shall not impair the operating functions of the crane. All guards and protective devices for nonelectrical components shall be made from metal.

3.5.2 Human factors. The characteristics of the crane shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic clothed, in accordance with SAE J833. Operator's controls shall operate within the individual's strength factors in MIL-STD-1472.

3.5.3 Sound level requirement. The sound level of the crane-shovel (excluding horns and warning alarms) shall be in accordance with MIL-STD-1474 Category D for steady-state noise when tested in accordance with 4.6.2.31. If the crane-shovel can be operated at a level that exceeds 85dB(A) the appropriate discussion of the noise hazard per

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MIL-STD-1474 (MANUALS) shall include the requirement for hearing protection, the noise level of the item, the distance from the item at which 85dB(A) will always be met, and what operational or system configuration situations will reduce the level to 85dB(A). Hazard signs, if required, shall say "CAUTION, HEARING PROTECTION REQUIRED".

3.6 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers, safety guards, and plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. The configuration of the crane-shovel shall provide maximum accessibility for removal of engine and drive train components. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked. Ease of maintenance provisions shall incorporate features insuring operating tool and lubricant component clearances, and shall facilitate maintenance and servicing operations at extremely low temperatures. Intricate locking devices and controls with threaded fastenings which can be overtorqued by personnel lacking feel through thick gloves or because of numbness of hands shall be avoided. Each operation shown on the following maintenance operations list shall be accomplished by one man using common tools and special tools furnished on the crane-shovel within the time limit specified when tested as specified in 4.6.2.5. After completion of the maintenance operations, the crane-shovel shall operate as specified herein.

Maintenance operations list:		Time allowed
(a)	Remove, replace, and adjust fan and accessory drive belts.	30 minutes
(b)	Remove, and replace alternator and readjust belt tension.	30 minutes
(c)	Remove, and replace electric starter.	30 minutes
(d)	Remove, clean, and replace all screens, strainers, and filter elements in the hydraulic systems.	30 minutes
(e)	Drain engine cooling system, remove and replace engine cooling system hoses and refill cooling system.	2 hours

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(f)	Drain, change filters, and refill engine oil system.	1 hour
(g)	Remove and replace fuel filter elements.	30 minutes
(h)	Disconnect battery cables, remove and replace batteries, and reconnect battery cable.	30 minutes
(i)	Remove and replace engine. * * A hoist and operator may be used for this operation.	4 hours each operation

3.6.1 Maintenance ratio. The ratio of manhours of maintenance required to the hours of operation performed shall not exceed .25 when tested as specified in 4.4.

3.6.2 Filling, draining, and checking provisions. The hydraulic-system reservoir and transmission housings shall be equipped with dipsticks to determine the fluid level. Other enclosures such as gear cases and reservoirs which contain lubricant or hydraulic oil, shall be equipped with dipsticks or with check plugs of not less than 1/2-inch pipe size to determine the fluid level. Dipsticks shall be etched, knurled, sand blasted, or phosphate coated to facilitate determining the liquid level. Each enclosure shall be equipped with means for filling and for draining. The drain shall be fitted with a magnetic drain plug having either tapered pipe threads or shoulder and gasket; or in lieu of the magnetic drain plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components except access covers. Each drain plug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the crane-shovel is in a level position, drainage of fluids shall be to the ground or into a 2-gallon bucket without draining on or over any part of the crane-shovel. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the filling means, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accessories and parts, except plates equipped with hand-operable, quick-disconnect fasteners, or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the crane-shovel. Extension hoses are permitted.

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3.7 Transportability.

3.7.1 Disassembly and reassembly. The following items shall be capable of disassembly and reassembly when tested as specified in 4.6.2.4, with a lifting device and no more than hand tools and special tools furnished with crane-shovel:

(a) Front-end attachments.
(b) Cab.
(c) Counterweight.
(d) Crawler base.
(e) Revolving superstructure.
(f) Swing circle or roller path.

3.7.2 Air transportability. The crane-shovel shall be disassembled in 6 hours and reassembled in 6 hours in conformance with the loading storage and weight limitations of attached figure 1 with not less than 5 inches clearance on both sides and a 6 inch clearance at top. A crane may be used to aid disassembly and assembly. Cutting and welding are not permitted.

3.7.3 Rail transportability. Each crane-shovel shall be disassembled in not more than 6 hours into units of a size that, when loaded on a flat car in accordance with Association of American Railroads "Rules Governing the Loading of Commodities on Open-Top Cars", will meet the Berne clearance dimensions shown in figure 2. The length of each unit shall not exceed the item length shown in figure 2. Reassembly shall require not more than 6 hours. Cutting and welding are not permitted.

3.8 Environmental and use requirements for nonwinterized crane.

3.8.1 Operating temperature requirements. The crane-shovel shall perform as specified herein in any ambient temperature from minus 25° F. to plus 125° F.

3.8.2 Storage temperature requirements. The crane shovel shall not be damaged by storage under the following conditions:

Air temperature
High - 160° F. Time period, 1 hour daily.
Low - minus 50° F. Time period, 6 hours. No solar radiation.

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3.8.3 Elevation conditions. The crane shovel shall perform as specified herein at any elevation from sea level (barometric pressure 29.92 inches mercury) and a maximum ambient temperature of 125° F. to 5,000 feet above sea level (barometric pressure 24.89 inches mercury) and a temperature of 107° F.

3.8.4 Use. The crane shall withstand the usage encountered in military operations such as operation over unimproved terrain, fording, and operation under all weather conditions specified herein without breakage or permanent deformation.

3.8.5 Mean-time-between failure. The "specified" mean-time-between failure (θ_o) shall be 68 hours when tested as specified in 4.6.2.34.

3.9 Performance.

3.9.1 Capacities.

3.9.1.1 Crane capacity. Each crane-shovel, when used as a crane with a 30-foot boom, shall have rated lifting loads of not less than those specified in table II when tested in accordance with 4.6.2.14.

Table II. Rated crane lifting capacities

Radius of load (ft)	Rated load (lb)
12	25,000
30	6,900

3.9.1.2 Shovel capacity. The shovel dipper capacity shall be not less than that specified in table III. The working heights, radii, and the digging depth of the shovel shall be not less than those specified in table III with the crane-shovel resting on firm level ground and the shovel boom positioned at a 45° angle.

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Table III. Shovel operating range

Shovel dipper capacity (cu yd)	Dumping		Cutting		Digging depth below floor level (ft)(in)	Cleanup radius at floor level (ft)(in)
	Height door open (ft)(in)	Radius max height (ft)(in)	Height (ft)(in)	Radius (ft)(in)		
3/4	14 6	22 0	21 0	26 6	5 6	17 6

3.9.1.3 Backhoe capacity. The backhoe bucket capacity shall be not less than that specified in table IV. The ground clearance at start of dump and the digging depth of the backhoe shall be not less than those shown in table IV with the crane-shovel resting on firm level ground.

Table IV. Backhoe Working Range and Bucket Width

Minimum bucket capacity	Minimum bucket width	Minimum ground clearance start of dump		Minimum digging depth	
Cu. yd.	In.	Ft.	In.	Ft.	In.
3/4	36	9	6	17	0

3.9.1.4 Clamshell and dragline capacity. Equipped with the basic boom, the crane-shovel shall be adaptable to and operable as a 3/4-yard-capacity clamshell and 3/4-yard-capacity dragline.

3.9.1.5 Crane capacity plate. All crane-shovels shall be provided with a crane-capacity plate mounted inside the cab in a location conspicuous to the operator. The plate shall be similar to and of the same material as specified in 3.22.1 and shall contain the following information (see 4.6.2.16):

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(a) Rated loads (see 6.9.3) in 5-foot increments for the full range of operating radii when using the basic boom length specified in 3.9.1.1 and for each 10-foot increment in boom lengths up to and including 60 feet. This plate shall also include capacity information for a 15-foot jib.
(b) Indication that the rated loads are gross hook loads at a radius measured from the center of rotation of the revolving superstructure and that the weights of any other external auxiliary lifting means are to be included as part of the hook-load.
(c) Statement that the rated loads are based on the crane standing on a supporting surface that is firm, level and uniform.
(d) Statement of deduction to be made from the rated loads for components of crane attachments such as jibs.

3.9.2 Operating characteristics. All operational functions for a given front-end attachment shall operate independently and simultaneously for maximum performance. These functions need not include travel.

3.9.2.1 Maneuverability. The crane-shovel shall be capable of traveling in a forward or backward direction, with the revolving superstructure in any position, and capable of being steered either to the right or left with controls from the operator's position. The completely equipped crane-shovel with applicable front-end attachment located either to the front or back of the crane-shovel shall conform to the following mobility requirements:

(a) Capability of ascending, descending and stopping on a 30 percent grade not less than 50 feet in length on smooth, firm, dry ground.
(b) Capability of negotiating a 15 percent side slope on smooth, firm, dry ground.
(c) Capability of travel on smooth, firm, dry ground at a speed not less than 0.75 miles per hour.
(d) Capability of fording and operation on hard-bottomed bodies of water up to 12 inches without evidence of damage or water leakage when tested as specified in 4.6.2.10.6.

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3.9.2.2 Ground bearing pressure. The ground bearing pressure shall be not greater than 11 pounds per square inch (psi).

3.9.3 Conditions. All requirements specified herein shall apply to crane-shovels with all lubricating and cooling systems full and all fuel tanks at least half full, unless otherwise specified.

3.10 Backward stability. The center of gravity from the axis of rotation shall be not farther than 70 percent of the radial distance from the axis of rotation to the tipping fulcrum in the least stable direction when the crane-shovel is resting on firm, level, and uniformly even supporting surface; in working order without load; and with the basic boom at the minimum working radius.

3.11 Design and construction. The design of the completely assembled unit shall be such that the maximum unit stress imposed by the required loading shall not exceed 60 percent of the yield point of any structural member when tested as specified in 4.6.2.13.

3.12 Superstructure. The superstructure shall consist essentially of a weather-tight cab enclosing the machinery and operator's position and shall be mounted on a full revolving platform. All operating machinery shall be mounted on and attached to the rotating platform. The rotating frame shall be a unit steel casting or heavy sections of structural steel welded together or a combination of cast steel and welded structural steel. The machinery shall be attached to the principal members of the platform frame. The principal member shall be designed to receive the front-end attachments (see 3.15) and pins shall be provided. Means shall be provided for retaining the pins. The pins shall be capable of being turned with common handtools for alinement. The machinery deck shall be sealed with boots or grommets where controls pass through the deck. Two clips shall be furnished and fastened on the front edge of the operating frame to provide connection points for the inner end of the piledriver lead catwalk (see 3.15.2.8). Grab handles shall be provided

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alongside the doors, and walks, and steps shall be located so as to facilitate safe access to the operator's seat and to components requiring frequent attention. Grab handles shall provide the hand clearance shown in figure 3. To facilitate repair, maintenance, and replacement of the equipment, folding hinged panels with labyrinth-type weather seals (see 6.5 and figure 4 and 5) shall also be provided for access to the engine compartment. A ladder shall be attached to the front of the superstructure cab to facilitate accessibility to the gantry and A-frame sheaves. When the fuel tank filler neck is more than 36 inches from the ground, an antiskid platform space not less than 15 inches square and 1/8-inch in thickness shall be located adjacent to and below the fuel tank filler neck to enable the operator to fuel the crane tank with 5-gallon cans or to aid in bulk fueling. This platform shall be hinged to the bottom of the superstructure to fold up and latch against the superstructure cab wall when not in use. The standing area shall permit the operator the full use of both arms. A catwalk shall be installed all around the superstructure except for the boom foot area. The catwalk shall be not less than 12 inches wide unless otherwise specified and shall be surfaced with open skid-proof grating. The catwalk shall support a concentrated load of not less than 400 pounds per square foot. The catwalk shall be bolted to the superstructure and may be removable or hinged to meet the width requirement (see 3.1.2). Access footholds or handholds to enable the operator to gain access to all platforms and catwalks shall be provided. Removable cast iron counterweights shall be provided. The complete counterweight shall be capable of removal in not more than 30 minutes by two men and set on a trailer or truck using only the lifting provisions on the crane-shovel. The complete counterweight shall be capable of replacement on the crane-shovel by two men in not more than 30 minutes using only the lifting provisions of the crane-shovel. The counterweight may serve as the rear catwalk providing the walking surface is skidproof and meets all requirements specified herein.

3.12.1 Functional requirements. With the engine operating at a speed not exceeding the governed speed, the crane superstructure shall meet the following requirements:

(a) A hoisting-drum line power of not less than 1,250,000 foot-pounds (ft-lb) per minute at minimum bare hoisting-drum line speed.
(b) A swing speed of not less than 3.5 revolutions per minute (rpm).

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(c) Bare hoisting-drum line speed between 160 and 180 feet per minute (fpm).
(d) Simultaneously lift and rotate not less than a 7813-lb load using a single part line after establishing the maximum safe operating radius.

3.12.2 Forward deck machinery protection. A sheet metal trough shall be provided to protect the forward machinery deck from mud and dirt scraped from the haul-in cable during dragline operations. The trough shall not interfere with any of the specified crane operations.

3.12.3 Cab. The sheet steel used for the cab shall be not less than 0.071 inch in thickness. The cab shall completely enclose the operator's seat, control levers and pedals, instrument panel, and all of the main-deck machinery. Hinged covers with slots for hoist cables shall be provided for weather protection of the cable drums. The arrangement of windows and windshield shall provide optimum vision for the operator and the seat, instruments, and controls shall be in accordance with SAE J833 and SAE J898. The cab shall have sufficient windows and glazed doors to permit 180 degree vision for the operator. Maximum unobstructed visibility of the work including the point sheaves of the basic boom at a 10-foot operating radius shall be visible to an operator varying from large to small in accordance with SAE J833 and wearing arctic clothing, seated upright. The seated operator shall also have 180 degree vision to all points projected on the ground at a horizontal radius of 12 feet and 20 degrees above horizontal for side window. All glass shall be held in place by rubber channels for replacement. All glazing shall be 1/4-inch, AS2, tempered, safety plate glass conforming to ANS Z26.1. Not less than 75 percent of the front window shall be opened and shall slide into the roof of the cab for storing. The window shall latch in both the open and closed position. Both the side and the rear doors shall be of the sliding type with locks that are accessible. The doors shall be constructed of metal and shall be equipped with all steel latches and locks. The seat as a unit shall be adjustable without tools. Vertical and horizontal seat adjustments shall be in maximum increments of 1 inch. The seat in the central position shall be located so that a man ranging in sizes as indicated in SAE J833 and clothed in arctic clothing is within reach of the control levers and pedals. Cab doors shall be held closed with a positive locking device and shall be provided with all-steel latches to prevent accidental opening and closing. Walking surfaces and steps inside the cab shall be of the antiskid type. The seat base shall be provided with a quick release to allow the seat to tilt sideways toward the machinery or forward to permit the operator's escape in case of emergency. Drainage provisions shall be made for any wells or depressions that are subject to collection of water.

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3.12.3.1 Cab heating. The cab shall be equipped with a heater conforming to MIL-H-3199, type II, capacity A, 24-volt rating, with a three-position switch including two speeds and on-off. The heater shall be mounted and positioned to permit routine maintenance and inspection and to allow for removal. The source of heat for the heater shall be from the superstructure engine. A ducting system shall be provided to direct cab heat toward the operator's feet and to distribute heat to the defroster. The ducting system shall include valves to permit the operator to select air distribution between the defroster duct and ducts for cab heat. Defroster shall be provided for defrosting the front windows. Ducting shall not interfere with operator performance and with ease of maintenance and shall be protected from damage by personnel. Controls for the heater shall be located so as to be within reach of the operator in his normal operating position. The heater ducts shall be so arranged to defrost not less than 75 percent of the front window area and shall be mounted and positioned to permit routine maintenance and to allow for quick removal.

3.12.4 Superstructure cab, disassembly. When necessary for transportability (see 3.7.2 and 3.7.3), the superstructure cab shall be capable of removal to reduce the size, weight, and shape of the cross section of the crane-shovel. The framing shall tie the side panels and main portion of the roof together as one unit, and the top corners shall be provided with lifting eyes to accommodate crane slings. The supporting members of the cab shall be bolted to the machinery deck with fasteners of sufficient strength and quantity so as to hold the cab securely in place under all operating conditions as specified herein. All free-standing edges of the removable and fixed section shall be provided with angle stiffeners and self-sealing, labyrinth-type weather seals as shown in figures 4 and 5 of a lapping construction so as to keep water out. All electrical, air, and hydraulic lines fastened to the walls of the cab shall be equipped with quick-disconnect, self-sealing couplings as needed at the separation line of the cab and the machinery deck.

3.12.5 A-frame. An A-frame or gantry shall be provided which shall be capable of withstanding stresses induced by lifting from the ground a standard military boom 60 feet long. The principal vertical compression and tension members of the A-frame shall be connected so that the operating stresses are transferred to the main structural members of the rotating frame of the superstructure. When operating with booms from 30 to 60 feet in length and the A-frame in the raised position, the centerline of the A-frame top sheave shaft shall be not less than 54-11/16 inches above

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the center of the boom foot pin when measured vertically nor less than 79 inches to the rear of the boom foot pin when measured horizontally. The A-frame sheaves having their grooves in a vertical plane shall be located laterally from a vertical plane through the longitudinal centerline of the boom so the angle between the centerline of the reeved boom hoist cable and a plane through the centerline of the sheave groove shall be not more than 6 degrees with the boom at 82 degrees from the horizontal (see 4.6.2.22). The location of all sheaves in the A-frame shall permit reeving through a pendant equalizer conforming to 3.15.1.2 and meeting all requirements specified herein. The gantry shall be of the folding or retractable type for overhead clearance or reducing cube during transportation.

3.12.6 Power load-lowering device. A power load-lowering device shall be provided on the primary drum to facilitate controlled load lowering. This device shall drive the hoist drum at a speed to produce a line speed 30 percent to 50 percent of the bare drum line hoisting speed as specified in 3.12.1. The device shall power the primary hoist drum in the direction to unspool the cable and shall be controlled by a clutch. The clutch shall be operated by a lever accessible to the operator in his seated position in the revolving superstructure. The power-down clutch shall have sufficient capacity to hold on a single line, without slipping, a load in pounds equal to the maximum line pull developed by the hoist drum (see 4.6.2.21). The clutch shall have sufficient capacity to overdrive the primary hoist drum brake with hook loads on a four-part line, or simulated loading of both maximum capacity and one-half maximum capacity with the brakes adjusted for holding a maximum capacity load on a four-part line. The boom angle shall not change and the brake lining temperature shall not exceed 600° F., or the maximum recommended by the manufacturer of the lining for the application, whichever is less, when tested in accordance with 4.6.2.20. Provisions shall be made to prevent damage to the gear or chain reduction when free spooling the drum. The controls for the load hoist and load lowering shall be entirely independent of the boom hoist and boom lowering controls.

3.12.7 Crane-boom hoist. The crane-boom hoist shall be independently controlled. The boom hoist, equipped with a positive holding device, shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for power lowering under the operator's control, and shall make free spooling of the boom cable impossible under any circumstance. The boom hoist shall raise and lower the 60-foot-long standard military boom under loadings and conditions specified in 3.12.1.

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It shall have sufficient rope capacity to allow the boom tip to be lowered to ground level with the crane-shovel resting on level ground and three wraps of wire rope retained on the drum. The boom hoist shall be equipped with a positive lock to facilitate holding the boom in any set position. The boom-hoist drum shall be arranged to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The drum shall spool the rope in even layers.

3.12.8 Drums. The drums shall be independently driven and convertible for use with any of the specified front-end operating equipment. The drums shall be arranged to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. Drums shall have sufficient rope capacity to accommodate the specified rope lengths (see 3.15.1.6.2). When clutches and brakes are not engaged, the complete drum assembly and mounting shall permit free rotation to allow gravity lowering of the hook block. The primary and secondary load hoist drums shall be operated by clutches controlled by levers so that once the clutch is engaged it shall be held in place by an overcenter lock (knuckle-in) on the control linkage. The clutches shall be unlocked by pushing or pulling on the control lever. In addition, the machinery operating the primary and secondary drums shall be arranged so that the drum clutches may be simultaneously engaged and, with the main engine-clutch disengaged, shall reverse the operating machinery by the lowering load under control of both drum brakes simultaneously for precision load lowering without the boom raising or lowering. Pitch diameters of drum or drum laggings including boom hoist drum shall be in accordance with SAE J881. The drums shall be provided with a socket equipped with a wedge for dead ending the wire rope. All drums shall be guarded to prevent the rope from leaving the drum during operation. Lubricating fittings or other form of lubricating devices shall be provided and so arranged to permit lubrication without unspooling the wire rope.

3.12.8.1 Hoist drum brakes. The drum brakes shall hold and slowly lower (without jerking or chattering) capacity loads supported by recommended reeving. Provision shall be made for locking the brakes in any position of the drum. Each drum brake (primary and secondary) shall hold the maximum load that the crane is capable of lifting on a single-part line at the minimum safe operating radius. The drum brakes shall be removable for lining replacement. All brake linings shall be riveted or bolted to the backing plate.

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3.12.8.2 Hydraulic lines. Hydraulic lines shall consist of tubing and hose. Hose shall be used where necessary to prevent failure due to vibration.

3.12.8.2.1 Hydraulic reservoir. Hydraulic reservoir shall be provided which shall have sufficient capacity to supply the required reserve fluid to the master cylinder to prevent entry of air into the hydraulic system. The reservoir shall be located at least 3-feet above the master cylinders.

3.12.8.3 Drum and swing clutches. Drum clutches shall be controlled by the operator while seated, shall be arranged to engage and disengage instantly under full engine torque, and shall require manual effort to engage. Drum and boom hoist clutches shall be of the friction or electromagnetic type, accessible for adjustment and repair, and large enough to eliminate slipping under 150 percent of rated load. Drum clutches shall withstand high-speed, continuous operation without overheating or excessive wear. The hoist clutches and travel clutch shall each individually be capable of stalling the torque converter and shall be located so as to be accessible for adjustment and repair. Swing clutches shall make not less than 3-1/2 cycles (a complete cycle of 90 degrees one way and 90 degrees back) per minute for not less than 1-1/2 hours without either fading or exceeding 500° F. with a 30-foot boom at an angle of 40 degrees above horizontal and with a load equal to 50 percent of maximum line pull. Clutches shall be controlled by mechanical linkage, air, or manual effort applied through hydraulic linkage or through variable pressure power hydraulic controls. When a power hydraulic control system is furnished the force applied to control levers or pedals shall be in direct proportion to the force exerted on clutches. All clutch facings shall be riveted or bolted to the backing plate. Hydraulic systems, if used to actuate the drum clutches, shall be capable of operation and storage without leaking or seepage of the hydraulic fluid. System component parts shall be compatible with one another and with the hydraulic fluid to prevent corrosion during storage or operation. The hydraulic master and slave cylinders shall have cylinder bodies and pistons of ferrous material. The cylinder cups shall conform to MIL-C-14055.

3.12.9 Rollers. The rotating frame shall be supported by large fixed rollers or by a live ball or roller ring. Rollers, balls, roller rings, and ball races shall be heat-treated hardened steel and shall be accurately machined. Fixed roller shall be fitted with lubricated bearings and shall be provided with grease fittings. The balls or rollers shall be accessible and shall be removable. When the holding-down function is

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performed by hook rollers, such rollers shall be either the dual-purpose type in which each set of rollers performs both the supporting and holding-down functions or the rollers may be of the holding-down type only. Hook rollers shall operate either singly or in pairs on tandem bogies and shall be provided with grease fittings. Hook rollers shall be provided with means for vertical or horizontal adjustment as required. The hook rollers shall be removable without removal of the hook-roller brackets from the revolving superstructure. The hook roller brackets shall not interfere with the removal of the superstructure from roller path and frame. The temperature of the supporting balls or rollers and hook rollers shall not exceed 175° F. under all operating conditions specified herein.

3.12.10 Swing-gear and lock and brake mechanisms. The revolving superstructure shall be actuated by a gear and pinion. The swing-gear pinion shall have accurately machined or ground teeth and shall be of alloy steel. The swing-gear pinion shaft shall operate in lubricated bronze or anti-friction bearings. A swing lock shall be provided to lock the revolving superstructure within plus or minus 2-1/2 degrees of any desired position except travel position. For travel, the crane boom shall be positioned to within 1/2 degree of the longitudinal centerline of the crawler frame. A swing brake shall also be provided to control the swing of the revolving superstructure and to hold the superstructure from swinging when the crane-shovel is working on a slope and shall require manual effort to engage. Both lock and swing-brake controls shall be located within reach of the operator while seated. The rotation of the rotating frame shall be controlled through friction-type or electromagnetic clutches designed for fast cycling, craning, and other operations, and with a clutch capable of transmitting not less than 130 percent of the maximum engine torque. The temperature of the swing clutches shall not exceed 500° F. and the clutch shall not fade when tested as specified in 4.6.2.19.

3.12.11 Power transmission. Unless otherwise specified (see 6.2), the engine shall be connected to the transmission through either friction clutch or friction disengaging clutch and fluid coupling. When specified (see 6.2), friction disengaging clutch and a hydraulic torque converter shall be furnished.

3.12.11.1 Clutch. The clutch shall be controlled from the operator's position, shall be arranged to engage and disengage instantly while the engine is running, and shall require manual effort to engage. The clutch shall be capable of transmitting not less than 150 percent of the maximum engine torque. The clutch shall be controlled by either mechanical linkage, air or manual effort applied through hydraulic linkage. Clutch facings shall be riveted or bolted to the backing plate.

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3.12.11.2 Torque converter. When specified, the hydraulic torque converter shall be a standard commercial design that has proven satisfactory in extensive crane-shovel use. The torque converter shall have a torque input rating of not less than the 150 percent of the maximum torque output of the engine. With an initial fluid temperature of not less than 180° F., in an ambient temperature of not less than 30° F., the torque converter shall absorb wide open throttle for a minimum of 30 seconds without exceeding 250° F. and shall return to a stabilized temperature within 15 minutes when tested in accordance with 4.6.2.35.

3.12.11.3 Drive chains. All drive chains shall be provided with a means of adjustment. This adjustment shall be made either by changing the center to center distance between the driving and driven shafts or by means of a roller or sprocket meshing with or riding on the chain. This roller or sprocket shall be located so the tensioning action shall be exerted on the exterior side of the slack side of the chain. The primary drive chain to the first sprocket shall conform to (a) below and shall be encased by an oiltight steel chain case, and run through an oil bath; or lubricant for the chain may be supplied by a circulating pump directing a spray of oil on top of the top chain span. The transmission chains shall conform to the following:

(a) Flat link plate roller chains conform to MIL-STD-421, type IRC, and limited to chain numbers 50-IRC-A, 50-3-IRC, 60-IRC-A, 60-3-IRC, 60-4-IRC, 60-5-IRC, 80-IRC-A, 80-4-IRC, 80-6-IRC, 100-IRC, 100-3-IRC, 100-4-IRC, 120-IRC, 120-2-IRC, 140-IRC, 140-2-IRC, 160-IRC, and 200-IRC.
(b) Offset link plate roller chains selected from table V conforming to MIL-STD-424.
(c) 60-degree inverted tooth silent chains conforming to MIL-STD-425 and limited to chain numbers SC616, SC1020, and SC1024.

Table V. Chain, Roller: Offset Link Plates, Power Transmission, Single Strand

Chain No.	1	2	3	4	5	6	7
Nominal pitch (in.)	6.000	5.000	4.500	4.073	3.500	3.067	2.000
Minimum ultimate strength (psi X 105).	3.6	2.5	1.93	1.41	1.22	0.90	0.5625

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3.12.12 Electrical system. The crane-shovel shall be equipped with a 24-volt, negative grounded, waterproof electrical system for lighting of the work areas at night, slave starting the engines of other vehicles, maintaining charge in the batteries, and starting the engine. The electrical system shall be in accordance with the following, except as specified herein:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b) MIL-STD-646.

All components shall be provided with the maximum protection from damage, shock and vibration through means of location and furnishing of resilient mounts and cushions. Sealed beam units and bulbs shall be accessible for replacement without distrubing the permanently installed housings and guards. Components shall be in accordance with table VI. Item quantities that are not specified shall be furnished as required for each crane.

Table VI. Vehicle Electrical System

Description	Requirement	Quantity
Battery, storage, 12 V, 6TN	MS35000	-
Holddown frame	MS53046-3	1 ea per 2 batteries
Holddown bolts	J-Bolts, 10-in. long, 3/8-in. diameter	5 ea per frame
Washers and wingnuts	MS15795-815 and MS35425-42	5 ea per frame
Clamp assembly	Tension type hook latches with strikers	-
Piano hinge	MS35829-C	-
Charging system, 60 amp at 28 volt	SAE J180 Commercial	1 ea
Low-tension cable	MIL-C-13486	-
Harness insulation	MIL-I-15126 and MIL-I-631	-
Loom	Polyethylene or SAE J562, oilproof loom	-
Cushioned support clamps	MS21322, MS21333, and MS21334	-
Circuit number bands	MS39020	-

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Table VI. Vehicle Electrical System (Cont'd)

Description	Requirement	Quantity
Horn	MS51074	1 ea
Friction connectors	MS27142 through MS27148 and MS33800	-
Connector lubricant	MIL-S-8660	-
Connector holder clip	Plastic or metal spring clip	1 ea per friction connector
Threaded plugs and receptacles	Waterproof military type	-
Terminal connectors	SAE J561 straight or side type spade	-
Insulator sleeves	Heat shrinkable tubing	-
Circuit breaker	MIL-C-13516	-
Slave receptacle	MS75058	-
Slave cable w/plugs on each end	MIL-C-13486, 240-in. length, and MS75057	1 ea, assy.
Temperature sending unit	MS24537	-
Pressure sending unit, 60 psi	MS24538	-
Pressure sending unit, 120 psi	MS24539	-
Fuel tank sending unit	Compatible with MS24544	-
Dome light	Commercial, 4 inch dia., elastomer mount	3 ea
Panel light	Commercial with Trade No. 313 bulb	3 ea, min.
Warning and indicator lights	MIL-L-3661/7 and MIL-L-3661/15	-
Battery-generator indicator	MS24532	1 ea
Pressure gage, 120 psi	MS24540	-
Pressure gage, 60 psi	MS24541	-
Temp. gage, 160 - 320° F.	MS24542	-
Temp. gage, 120 - 240° F.	MS24543	-
Fuel gage	MS24544	1 ea
Tachometer, hourmeter	MS35916	1 ea
Rotary switch	MS39060	-
Toggle switch	MS39061	-
Push button starter switch	Commercial water-proof, 15 amp. min	1 ea

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Table VI. Vehicle Electrical System (Cont'd)

Description	Requirement	Quantity
Work lights	MIL-H-13071	4 ea
Sealed beam units, flood	MS17165-4578	4 ea
Starter	MIL-S-3785	1 ea
Metal conduit	Commercial thinwall electrical type	-
Conduit clips	Commercial type	1 ea per 4 ft span

3.12.12.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition as specified in 3.12.12.1.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.12.1.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.12.12.2 Electromagnetic interference. The electromagnetic interference emission of the crane-shovel shall conform to MIL-STD-461, Tactical and Special Purpose Vehicles and Engine Driven Equipment.

3.12.12.3 Wiring. The wiring of the crane shall be fabricated and installed to provide a reliable electrical system. Rigid conduit shall be furnished for all circuits to within 12 inches of components, with the exception of the battery and slave cables. Loom or harness insulation shall be furnished from the end of the conduit to those components connecting two or more conduits. Conduit, clips, loom, grommets, and

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cushioned support clamps shall prevent whipping, chafing, or other mechanical damage to the wiring or harness. Wiring shall be routed within the framing members whenever possible. Unless protected by conduit or loom where two or more wires follow the same routing, the wires shall be formed into harnesses by binding together with one-half overlapping turns of 0.010-inch-thick, 1-inch-wide tape or with insulation. The wiring shall facilitate defective wire replacement from bulk supply. Battery and slave cables shall be of adequate cross section to satisfy the SAE starter voltage drop in SAE J541 for heavy duty diesel use, 24 volts, and maximum current requirements. These cables shall not be spliced. The size of other wiring shall also be based on voltage drop and current required plus 15 percent allowance increase for deterioration and in no instance shall the cable be smaller than A.W.G. 14. All circuits shall be identified by number and color coding, on circuit number bands, furnished near each connector. Color coding shall be limited to these bands. As applicable to crane-shovels, the number and color coding shall follow the system in accordance with MIL-STD-646. Friction connectors (MS27142 through MS27148 for wire and MS33800 for components) shall be lubricated prior to assembly connection. Each friction connector assembly shall be secured to the crane-shovel with a connector clip. Terminal connectors shall be of the type equipped with either heat shrinkable or integral tubing sleeves. Removal of major assemblies such as engine, cab, and instrument panel cluster shall be accomplished by disconnecting one or more threaded plug and receptacle for all 24-volt circuits in each assembly with the exception of battery, starter and slave cables. When circuit quantities exceed 12 in number, one plug and receptacle shall be furnished for each 12 circuits, or fraction thereof, and the multiple disconnect points shall be in the same location. All circuits shall be protected by circuit breakers.

3.12.12.4 Lights.

3.12.12.4.1 Floodlights. For illuminating the working area, the floodlights shall conform to table VI, size PAR-46. Each lamp shall be provided with a housing conforming to MIL-H-13071 and the lamp housing shall be attached to the crane shovel superstructure as specified herein by a mounting bracket conforming to and having the adjusting features as specified in MIL-H-13071. The housings shall be mounted in a position to permit replacement of the lamp. Lamp housings shall not extend beyond the side of the revolving superstructure unless provision is made for folding the lamps clear of the sides of the superstructure for transport. The floodlights shall be located so that lenses will not be seen by the operator in his normal operating position and shall illuminate the working area

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directly in front of the operator. One master light switch shall be furnished on the instrument or lighting panel to prevent operation of any light during blackout conditions, with the exception of engine indicator lights. The master switch shall be equipped with a red, spring loaded hinged cover to prevent inadvertent actuation by the operator. Four toggle switches, each with red indicator light, shall be furnished and banked near the master switch providing selective actuation of top-mounted floodlights, skirt-mounted floodlights, ceiling cab lights and instrument panel lights, whenever the master light switch is in the "ON" position. Each switch shall be identified as to function and position.

3.12.12.4.1.1 Top-mounted floodlights. Two floodlights, one on each side of boom hoist cables, shall be mounted at the top front edge of the cab.

3.12.12.4.1.2 Skirt-mounted floodlights. Two floodlights, one on each side of the superstructure skirt shall be mounted as near the side and bottom of the cab skirt as practicable but not higher than the mid-point between the bottom of the cab skirt and the bottom of the operator's window.

3.12.12.4.2 Interior cab lights. The cab shall have three interior ceiling lights effectively located with plastic lens not less than 4 inches in diameter. The light shall be of the flush-mount, waterproof type having either a flexible elastomer housing or an elastomer ring. Each light shall be equipped with a single contact, bayonet type bulb, Trade No. 1683. The overall depth of the light fixture from the face of the light to the ceiling shall not exceed 3 inches.

3.12.12.5 Warning horn. The revolving superstructure shall be equipped with a warning horn conforming to MS51074. The horn shall be mounted in a protected location on the exterior of the revolving superstructure so that it does not extend beyond the sides of the cab. The horn shall be operated by a heavy-duty-pushbutton (normally open) switch mounted on the swing lever of the revolving superstructure.

3.12.12.6 Trouble lamp. A hand-grip type, portable, trouble light shall be mounted in the cab adjacent to operator's door. The light shall be equipped with a Trade No. 4505, sealed-beam spotlight and furnished with a connecting self-coiling cable not less than 25 feet long. The trouble light shall have a hand grip handle incorporating a thumb or finger on-off switch. The trouble-light mounting bracket shall be of

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the cushioned spring-over-center type holding and enclosing the sealed beam unit to protect it from damage during all operations specified herein.

3.12.12.7 Instrument panel. An instrument panel containing all switches, gages and instruments, except hourmeter-tachometer, for complete control of the engine and lights shall be located within reach and visibility of the operator in his normal operating position. All switches, gages and instruments shall be identified as to nomenclature and positions, as a minimum requirement. Identification plates shall be securely fastened to the panel to prevent breakage, bending or other defacement and the material shall conform to 3.22.1. All instruments and indicators shall be installed assuring a metal-to-metal return ground from the component to the battery. The instrument cluster group shall be removable from the front to permit access to all wiring and units. Panel lights shall be furnished for visual reading of all instruments, indicators and identification plates by the operator. Lighting intensity on the face of each component shall be not less than 30 foot-candles. The instruments and gages shall be mounted within gage functioning tolerance. All warning and indicator lights shall be banked horizontally on the instrument panel. All primary instruments, indicators and controls for diesel-engines as defined in SAE J209 shall be furnished and shall be not less than the following:

Item	Requirement
Battery-generator indicator	MS24532
Fuel gage	MS24544
Heat indicator (engine coolant)	
Indicator	MS24543
Transmitter	MS24537
High temperature warning light	MIL-L-3661/14 Red lens
Heat indicator (torque converter and engine)	
Indicator	MS24543
Transmitter	MS24537
High temperature warning light	MIL-L-3661/14 Red lens
Oil pressure gages	
Indicator	MS24541 or MS24540
Transmitter	MS24538 or MS24539
Low-pressure warning light	MIL-L-3661/14 Red lens
Hourmeter-tachometer	MS35916

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The hourmeter-tachometer may be mounted on the engine in a location that is visible to the operator in his normal operating position. The hourmeter shall have a totalizing mechanism to show not less than 9,999 hours to register the number of hours of engine operation. The tachometer shall be accurate within plus or minus 5 percent of actual engine speed. All panel wiring shall be routed to eliminate a tangle of crossed wiring behind the instrument panel. Electrical wires shall be bundled into a harness assembly routed along an edge of the instrument panel with component connection lines parallel to each other line.

3.12.13 Operator controls. The operator's controls shall be located and installed within the maximum areas in accordance with SAE J898.

3.12.13.1 Crane controls. The crane, and other operating attachments shall be controlled by the least possible number of hand levers and foot pedals necessary for efficient operation. The levers and pedals shall be within reach of the operator when seated and shall not obstruct the operator's view of the work. All brakes shall permit locking and unlocking by foot action alone but shall not become locked without the operator's intent regardless of the operator's size, skill, or training. For ease of operation, the pedals shall rise from the depressed position in a backward as well as vertical movement. Hand- and foot-operated controls shall be in accordance with SAE J983. Latches on control levers shall not cause delay in operation. Instruction plates conforming to 3.22.1 shall be located on or near each control lever or pedal to indicate the function. The neutral and operating positions shall be shown by arrows and words. When the same lever controls more than one function, all functions shall be listed. The method of lettering and the method of fastening the plate to the panel board shall be in accordance with 3.22.1.

3.12.13.2 Engine hand controls. Mechanical push-pull remote controls shall be of the waterproof, antifriction type. As a minimum, all exposed parts shall be fabricated of corrosive-resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding in the mounts or cable. Bend radii shall be as recommended by the cable manufacturers plus an increase in radii of 20 percent. The as-installed mechanical efficiency of each installation shall be not less than 71 percent and the no-load friction shall not exceed 5 ounces per foot. The continuous operation load rating, in both tension and compression, shall be not less than 150 pound rating for crane-shovel controls and 200 pounds rating for engine controls and other controls.

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3.12.13.3 Accumulator. The accumulator, when furnished in the hydraulic control system, shall be of the piston type and charged with nitrogen gas. The pressure in the accumulator after completion of the tests specified in 4.6.2.3 shall be not less than it was at the start of the test. The accumulator, when part of a hydraulic control system incorporating an engine driven pump, shall be of sufficient capacity to assure control of a clutch for not less than 15 full cycles of engagement and release without engine power.

3.13 Engine. The diesel engine, including all systems, components, accessories, and auxiliaries, shall conform to MIL-E-11276, except as otherwise specified herein.

3.13.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the crane-shovel under the conditions specified herein shall not require horsepower or speed in excess of the intermittent brake horsepower rating of the engine at the applicable speed as established by the engine Qualified Products List. The governed speed shall not exceed the highest speed at which the engine was previously qualified by the engine test for intermittent operation, and at no point on the horsepower curve shall smoke samples show a reading of more than 5 on a Robert Bosch EFAW 68 smoke meter. The maximum governed speed shall be between 120 and 150 percent of the peak torque speed. Engine, with accessories installed, shall be capable of developing intermittent brake horsepower at full load governed speed of not less than 275 percent of the drum line power specified in 3.12.1(a). The engine shall have a maximum no load governed speed appropriate for the torque converter used but not more than 2600 rpm.

3.13.2 Operation condition. All systems, including lubrication system, induction system, and coolant system, shall perform without leakage or malfunction when the crane-shovel is negotiating a 30 percent grade or a 15 percent side slope when tested as specified in 4.6.2.10.

3.13.3 Lubrication system. Except for the oil filter requirements, the lubrication system shall be furnished in accordance with MIL-E-11276 and shall be tested in accordance with 4.6.2.32. Oil consumption limits specified in table II, test method 1500 of MIL-STD-1400, schedule III, shall not be exceeded. The oil shall be filtered by truck-type, diesel-engine oil filters. The lubricating oil filters shall conform to MIL-E-11276 and shall be of the full flow type. The filters shall be located in an accessible location that will permit changing of the filter elements. The external oil lines leading to and from the filters shall be

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of flexible hose with reusable fittings. The hose shall be constructed of corrosion resistant steel wire with a tube of tetraflouroethylene. The lines shall be compatible with lines and fittings conforming to SAE J40, SAE J40R2, type E or F. The lines shall be identified by markers conforming to MS39020 labled "ENG OIL" near each fitting.

3.13.4 Fuel system. Except for the fuel tank requirements, the fuel system shall be furnished in accordance with MIL-E-11276. A fuel tank of welded aluminum construction, 0.125 inch in thickness, conforming to DOT Safety Regulations and to MIL-T-46786, type III, shall be furnished. The fuel tank shall have a capacity of not less than 50 gallons and shall be located under the machinery deck. The tank shall be equipped with a drain outlet and threaded plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the tank directly to the ground. The fill-pipe shall be marked "DIESEL FUEL ONLY" on a plate conforming to 3.22.2. The tank filling components shall include fill-pipe, strainer, safety cap, and external vent. The fill-pipe shall extend through the outside of the cab wall in a protected and accessible location so that the tank can be filled from a 5-gallon container conforming to MIL-C-1283. The fill-pipe shall be equipped with a safety cap which is captive chained with swivel to the strainer and fill-pipe. The strainer proper shall be not less than 4 inches ID and shall be made of AISI 300 series corrosion resistant steel screening not less than 80 mesh, protected in such a manner as to prevent contact with the filling nozzle. The strainer shall be externally protected from damage due to surging of fuel. The fill-pipe, strainer, and cap shall conform to DOT regulations and shall permit filling the tank at the rate of 50 gallons per minute without spilling. The engine fuel intake line shall be in the center of the tank. The fuel pickup point shall be not less than 1/2 inch above the bottom of the tank and shall be located on the longitudinal center of the tank to prevent starving of the engine on grades and slopes. The tank shall be baffled to prevent surging of the fuel. The fuel lines from the tanks to the final filter on the engine shall be of flexible hose with reusable fittings, all conforming to SAE J40, SAE 40R2, type E or F. The lines shall be routed in protected locations not subject to chafing and shall be supported by cushioned support clamps. The lines shall be identified by markers conforming to MS39020 labeled "ENG FUEL" near each fitting. There shall be no evidence of raw fuel leakage throughout test specified in 4.6.2.10. The mounting of the fuel tank shall be as recommended by the fuel tank manufacturer to prevent shifting, damage and fatigue as a result of travel and transport as specified herein. Brackets and fuel tank straps shall be insulated from the tank by a non-fuel absorbent cushioning material.

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3.13.5 Cold starting aids. Unless otherwise specified (see 6.2) engines shall be provided with cold starting aids either conforming to MIL-E-52649, type III, size as recommended by the engine manufacturer, or of the glow plug type, suitable for cold temperature engine starting for meeting the requirements specified in 3.8.1.

3.13.6 Cooling system. A cooling system conforming to MIL-E-11276, except as otherwise specified herein, shall be furnished. When the loaded crane is operated at 5,000 feet above sea level or in an ambient temperature of not less than 125° F., the cooling system shall maintain the coolant in the top tank at a temperature not to exceed 210° F. The control of the engine operating temperature shall be accomplished by a bleed-hole-type engine thermostat or a full-blocking-type engine thermostat that prevents coolant flow through the radiator core when the thermostat is closed. The radiator shall be of the three piece bolted construction type; top-header tank, core assembly, and bottom-header tank. The radiator shall incorporate a de-aeration system that shall function to eliminate air from the engine coolant and shall permit operation of the crane-shovel on grades and side slopes, as specified herein, without sucking air into the tubes of the core. The cooling system shall include sectionalized, removable shrouds. The shrouds shall be fabricated of asbestos-filled polypropylene plastic or of aluminum sheet metal and the baffle seals shall be of flexible, elastomer material. Fan guards of the sectionalized, removable type shall be required. The radiator shall be secured by two metal braces (one on each side). These braces shall be fastened to the radiator shell approximately halfway between the top and bottom tanks and shall extend approximately 45 degrees to a point where the other end shall be fastened to the machinery deck. These braces shall have sufficient rigidity to prevent vibration damage to the radiator or the radiator connections at all engine speeds. The cooling system shall be pressurized and shall be furnished with a large size filler neck and cap for either 7 or 13 psi systems in accordance with SAE J633. A cooling system filter shall be provided that is matched to the coolant capacity of the engine. The filter shall retard electrolytic corrosion by providing zinc anodes. The element shall contain ion exchange resins to treat the coolant and maintain a correct acid-alkali condition. All chemicals in the filter shall be compatible with all types of coolants and shall be noncorrosive to the metals used in the cooling system in accordance with ASTM D1384. The mechanical filter of the element shall prevent passage of any particle larger than 20-micron size. The inlet and outlet of the filter shall include shutoff valves. The filter shall be installed in accordance with the recommendations of the filter manufacturer.

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3.13.7 Air induction system. The engine shall be equipped with a dry-type air cleaner. The air cleaner shall conform to MIL-A-52363, class III. An air cleaner restriction indicator, conforming to MS53063, shall be installed inside of the operator's compartment near the instrument panel and visible to the operator to inform the operator when the air cleaner requires servicing. The intake vacuum of the installed system shall not exceed maximum vacuum recommended by the engine manufacturer.

3.13.8 Engine exhaust system. An engine exhaust system conforming to MIL-E-11276, except as specified herein, shall be furnished. The exhaust system shall expel gases to the rear of the crane-shovel at a height of not less than 10 feet above the ground. The exhaust pipe configuration shall prevent entrance of windblown precipitation into the cylinders of the engine. The exhaust muffler, if other than horizontal, shall terminate with a curved stack pipe. The curved stack pipe shall be of the truck-type having at least a 14-inch radius of curvature and a beveled undercut to protect the system from the weather. The aiming of the exhaust pipe shall be angled between 0 to 90 degrees towards the side from the rear of the crane-shovel and shall direct the gases away from personnel standing on the ground. The exhaust pipes, muffler, and stack pipe, shall be inherently heat and corrosion resistant or shall be protected on both the internal and external surfaces to resist the effects of heat and corrosion. The pipes shall be protected from heat, rust and discoloration by either a ceramic coating fused to both the interior and exterior of the pipe, or use of corrosion resistant steel. The ceramic coating shall provide thermal shock resistance at all temperatures up to 1650° F. The mounting bracket assembly shall be installed to withstand a force of at least 300 pounds, imposed in any vertical and horizontal direction without exceeding 1-1/2-inch displacement and without permanent deformation, damage, or loss of sealing of any component. The method of mounting, bracing, and clamping of the exhaust components to the bracket assembly shall prevent damage from heat expansion and from severe vibration occasioned by cross-country travel. The exhaust back pressure of the installed system shall not exceed the maximum recommended by the engine manufacturer. At least one flexible section of corrosion-resistant steel bellows shall be provided between the engine and the muffler. A muffler shall be furnished and shall have an internal volume of not less than 4 times the cubic inch displacement of the engine.

3.13.9 Governor. The engine shall be controlled by a tail shaft governor in addition to the engine overspeed governor. Selected tail shaft governor settings through the operating range shall be provided by a variable-speed-governor control lever which shall be accessible to the operator in the seated operating position.

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3.13.10 Starting system. The engine starting system shall be as recommended by the engine and the starter manufacturers for this application. The starting system shall include a starter switch, auxiliary control solenoids, engine speed disconnect, main starter solenoid, starter motor, and fly wheel drive mechanism. The starter switch shall be of the waterproof type, rated at not less than 15 ampere inductive load. The starter switch shall actuate a 24-volt auxiliary control relay which shall actuate the main starter solenoid and starter motor. The engine speed disconnect shall prevent starter engagement whenever the engine is running. The disconnect switch shall conform to MS90530-1 and shall be operated by fuel or oil pressure. The starter motor shall conform to MIL-S-3785, type II, grade A, class 1, heavy duty. The starting system shall withstand at least one discharge of the batteries from full charge (normally 24-volts) to the one half voltage of not more than 12-volt battery potential under the starter cranking load. Procedure for starter engagement periods shall be in accordance with the starter manufacturer's recommendations.

3.13.11 Charging system. An engine-driven battery-charging system shall be furnished. The 60-ampere, a-c generator (alternator), conforming to SAE J180, shall maintain battery charge and shall supply electrical energy to operate and illuminate the electrical energy to operate and illuminate the electrical system components specified herein. The charging systems shall be of the low cut-in type that provides not less than 25 amperes hot charging current at engine idle speed with the lowest possible alternator speed. The alternator speed range shall be in accordance with the recommendations of the alternator manufacturer and shall not exceed 6,000 rpm, maximum. The alternator shall be mounted on two fixed mounts and one adjustable mount in accordance with SAE J180. All alternator mounting bolts shall be not less than SAE grade No. 5 of a size not less than 1/2 inch. Hardened steel flat washers, not less than 1/8 inch thick, shall be furnished under all bolt heads, nuts, and between the adjusting lug and the adjusting strap, to prevent overstressing of the mounting surfaces. The alternator adjustable mounting bracket shall provide movement for adjusting belt tension and shall resist belt tension forces to minimize potential loss of recommended belt tension. The alternator fixed mount axis line shall be within a range of 20 to 60 degrees from the belt tension resultant load direction, as measured from the alternator armature shaft centerline. The fixed mounting bracket shall be in compression as a result of the load applied by the belts. A protective relay assembly, electrically matched to the A-C generator, shall be provided to protect the crane-shovel electrical system from damage when (1) incorrect polarity is

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applied through the slave receptacle to the batteries, (2) the crane batteries are incorrectly installed, and (3) arc welding for any maintenance or repair is performed on the crane. Fuses are not acceptable. Protective diodes, if used within the system, shall be installed in a manner which permit removal and replacement without the use of tools. The diodes shall be of the preferred types, as specified in MIL-STD-701, having a minimum amperage rating of not less than 150 percent of the maximum current flow and a peak inverse voltage (PIV) rating of not less than 1200 volts.

3.13.12 Storage batteries. Storage batteries shall conform to MS35000-3, except that electrolyte shall be furnished. The electrolyte for each crane-shovel shall be separately packaged in accordance with 5.1. The batteries shall be connected to provide 24 volts and shall be negative grounded.

3.13.12.1 Battery container. A battery container shall be provided for the batteries. The container shall be fabricated of non-corrosive sheet metal not less than 0.060 inch nominal thickness and the interior of the container shall have a finish in accordance with MS53046. The enclosure shall (1) hold the batteries level when the crane-shovel is on level ground, (2) include drains for water or electrolyte spilled within the enclosure, (3) provide gas venting to the outside, at or near the top of the enclosure, and (4) provide accessibility for servicing and removal of the batteries. Drainage shall be to the ground without draining over other parts or components. Clear vision shall be provided for checking of the electrolyte level in each cell of the batteries. Each pair of batteries shall be fastened to the enclosure to prevent damage from movement and vibration. Battery holddown frames shall conform to MS53046-3 and shall be installed with a total of 5, 3/8 inch 10-inch-long "J"-bolts, washers conforming to MS15795-815, and wing nuts conforming to MS35425-42. The size of the enclosure shall provide not less than 1-inch clearance of each side and top of a pair of 6TN battery cases as shown in MS35000. The cover shall overlap the sides and ends of the enclosure, and the cover shall be either removable or hinged to open to an angle greater than 90 degrees to permit vertical removal of the batteries. A spring loaded, detent brace shall be furnished at each side to hold the cover open when hinged. Tension type latches with strikers shall keep the cover closed. Battery containers that are located in a position providing a stepping surface, shall support the maximum weight of personnel stated in SAE J833 without permanent deformation. A pictorial battery hookup plate, in accordance with 3.22.1 shall be furnished on or near the battery container. The positive and negative terminals shall be identified.

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3.13.13 Emergency starting and charging receptacle and cable. A slave receptacle conforming to MS75058 shall be provided to facilitate electrical connections between the batteries and an external electrical power source for emergency starting of the engine and charging of the batteries. The receptacle shall be recessed in the wall of the superstructure so the plug-in face of the receptacle is flush with the outside face of the wall and shall be accessible to a man standing on the ground. The receptacle shall be located as near the batteries as possible and shall be connected into the battery circuit. A 20-foot-long cable shall be furnished for each crane shovel. The cable shall be two-conductor sheath cable conforming to MIL-C-13486, type I, class B equipped with an MS75057 slave plug on each end. The emergency starting and charging receptacle and slave cable shall conform to 3.12.12 and shall perform as specified herein when tested as specified in 4.6.2.28.1.2. Provision shall be made for carrying the cable within the superstructure cab.

3.14 Crawler mounting.

3.14.1 Crawler base. The crawler base shall be a rib steel casting or shall be rigidly constructed of heavy steel shapes, fitted and welded, and shall be constructed to prevent permanent distortion or excessive yielding in traveling across sharp depressions and uneven ground. The swing gear and turntable roller path shall be steel and shall be rigidly attached to or integral with the base. The swing gear shall have accurately machined or ground internal or external teeth. The turntable roller or shear ball path shall have an accurately machined or ground bearing surface, and shall be of sufficient size and strength to support the revolving superstructure under all conditions without distortion. Where the center pin performs the holddown function to prevent tipping, the pin shall be of high-strength steel and shall have all bearing surfaces accurately machined or ground to size and shall be provided with adjusting devices readily accessible from the rotating frame, or the center pin may be of a separate flanged casting, or forging, bolted to the underframe. Where hook rollers perform the holddown function, the lower member of the center pivot bearing shall be integral with, shrunk into, or welded to the crawler base and shall be steel with smooth bearing surfaces. The center shaft or pin shall operate in bronze or antifriction bearings.

3.14.2 Track assembly. The crawler tracks shall be of a width and length suitable for the crane-shovel; shall comply with ground bearing pressure requirement specified in 3.9.2.2; shall be constructed for

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operation in loose dirt, mud, rocks, and snow; shall be provided with suitable adjustments for maintaining proper tension in the track belts; and shall be designed so that the track belts cannot be accidentally disengaged. The tract shoes shall be of wear-resisting steel and of the flat self-cleaning design. The track-connecting pins shall be of steel and of sufficient size, hardness, and strength for the purpose intended and shall be readily removed and replaced. The tracks shall be supported on a sufficient number of steel rollers in such a manner so as to distribute the weight of the entire crane-shovel uniformly over the ground for the entire length of the crawlers, without depending upon the tension in the track for support. All track assembly shafts and rollers shall be provided with guarded grease fittings. Where practicable, grease fittings shall be located so as to be accessible from outside the crawlers and lower base.

3.14.3 Side frames. The crawler track side frames shall be steel castings or welded structural steel members. When side frames are not used, large track rollers shall be used with not less than four axles supporting the lower base.

3.14.4 Crawler drive. The crawler shall be driven by suitable machine cut gears or roller chains. The travel mechanism shall have positive steering, by means of independently operated friction brakes and adequate clutches or positive locking lugs engaged by sliding jaw clutch member, controlled from the operator's cab. The crawler clutches shall be of adequate size to permit constant operation without overheating. Brakes or other locking means shall be provided to hold the crane-shovel stationary during the working cycle while the crane-shovel is on level grade, or any grade at which it will operate satisfactorily and, to prevent the crane-shovel from accidentally rolling down a 40 percent grade. Brakes that can be applied at any time regardless of the position of the dogs on the propulsion drive shall be furnished on the travel mechanism. It shall be possible to apply the brakes on both tracks at the same time when required. Jaw clutches, when furnished, shall have a sufficient number of teeth to be readily engaged and to transmit maximum traction power without undue wear. One jaw clutch must be engaged at all times. The travel gears and bearings shall be protected against the entrance of dirt or other foreign matter. The propelling mechanism shall be provided with a positive traction lock or brake for locking the crawlers. The propelling shaft shall be so arranged that it is not one or a part of the supporing axles. All shafts shall be properly bushed or provided with antifriction bearings and ball-check grease fittings.

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3.14.5 Axle. Each axle or crossmember, when used, shall be either of cast, forged, or structural steel. When more than one-piece structural steel axles are furnished, the edges to be joined shall be alined and shall have a continuous weld so that the requisite weld penetration for transference of maximum design stress through the base metal juncture shall be provided.

3.15 Front-end attachments. The revolving superstructure of each crane-shovel shall be equipped with all necessary equipment to permit operation of the crane-shovel as a crane (boom with or without a jib), clamshell, dragline, piledriver, shovel, backhoe and crane with wrecking ball. Tools, equipment, parts and material which are not a component part of the crane-shovel as specified herein and are required for operation of front end attachments shall be furnished with each front end attachment.

3.15.1 Primary front-end attachments. Unless otherwise specified the following attachments shall be furnished with each crane-shovel and such attachments shall be complete with all necessary parts, including wire rope required for satisfactory functioning.

3.15.1.1 Crane boom. The basic boom shall conform to MIL-B-52203 and shall be 30 feet long. Boom lengths up to 60-feet shall be accomplished by using 10-foot extensions (see 3.15.2.1).

3.15.1.2 Pendant equalizer assembly. A pendant equalizer assembly shall be furnished. Stresses in the pendant equalizer shall not exceed 60 percent of the yield point under all operating conditions specified herein and shall be tested as specified in 4.6.2.13.

3.15.1.2.1 Sheaves. The pitch diameter of load bearing sheaves shall be in accordance with SAE J881. Flanges shall extend beyond the rope grooves not less than 1.75 times the diameter of the rope and shall have a hardness of from 255 to 302 Brinell. All sheaves shall be guarded to prevent the rope from leaving the sheave groove during operation. The sheave pins shall be of medium carbon or alloy steel and shall be of sufficient size and strength to support the sheaves under maximum operating conditions.

3.15.1.2.2 Sheave shaft. Sheave shafts shall be steel ASTM A576, grade 1040-1045 and a hardness of 229-277 Brinell. Means of retaining the sheave shaft in the assembly shall be provided.

3.15.1.2.3 Clevis pin. Clevis pins shall be furnished for attaching pendant equalizer assembly to the open swaged socket of the pendant cable assembly, MIL-B-52203. Clevis pins shall be provided with means of retaining the pin and shall be steel ASTM A576, grade 1040-1045 and a hardness of 240-265 Brinell.

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3.15.1.3 Boom safety stop. A telescopic boom stop shall be furnished in order to provide a safeguard against overtopping of the boom and shall be installed so as to limit the maximum boom angle to 82 degrees above the horizontal plane. The complete boom stop with supporting attachments shall be designed with sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock-absorbing action and shall remain assembled when the boom point is lowered to the ground. The boom stop shall be attached to the boom without bolting to or welding to any component part of the boom and shall be removable with common tools. Under all operating conditions, as the boom is raised from the ground, the boom stop shall telescope freely without sticking, binding or bending while the boom is raised to an angle of 82 degrees with the horizontal.

3.15.1.4 Boom angle indicator and chart. An indicator which will show the position of the boom at all times shall be provided. The indicator shall be of the heavy-pendulum type swinging on a double-sealed (sealed-for-life) antifriction bearing. The pointer shall indicate the angle of the boom with the horizontal plane. The indicator shall be provided with means of attaching to the standard military boom without cutting, welding or drilling the boom. The indicator shall be located on the boom, near and visible to the operator in his normal operating position. An angular radius relationship chart shall be furnished and mounted in the cab near the operator. The chart shall show angles and radius with boom length up to 60 feet with a statement of additions to be made when using the boom jib. The chart shall be of the same material as the plate specified in 3.22.1. The chart may be combined with crane capacity chart specified in 3.9.1.5.

3.15.1.4.1 Overload indicating system. When specified (see 6.2), an overload indicating system shall be furnished in accordance with SAE J159 and as specified herein. This system shall automatically indicate and warn the operator when an overload condition exists that exceeds the rated loads through the use of lights, dials, and an audible signal mounted within the operator's view within the cab. The audible and light signals shall cease and dials shall indicate a safe loading condition when corrective action has been taken by the operator. Any components of the system mounted on the military standard boom shall be installed without cutting, welding, or drilling the boom. The indicating system may utilize the 24 volt electrical system of the superstructure. The electrical system shall be waterproof and shall operate under all temperatures and use conditions specified herein without any malfunction.

3.15.1.5 Crane hook block. The crane hook block shall conform to GGG-B-485 except that the hook block shall have a safe working capacity required to meet the crane shovel requirements. The hook block shall have a proof test load capacity of twice the safe working load capacity and shall contain two sheaves. The overall length shall be not more than that specified for the size 20 and the gross weight and hook opening not less than that specified for the size 10 of GGG-B-485. Lubrication fittings shall conform to 3.17.

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3.15.1.6 Rope. All rope except tag line and dipper trip lines shall be wire rope conforming to RR-W-410, type I, class 2, 6 x 19, construction 3, IWRC improved plow steel, uncoated, right regular lay, and shall be preformed. The breaking strength of ropes shall be not less than the wire rope strength factors conforming to SAE J959. The hook block load rope shall be not less than 5/8-inch diameter, and boom hoist shall be not less than 1/2-inch diameter.

3.15.1.6.1 Rope sockets. All ropes, except taglines, and dipper trip lines, shall have ends attached to drum laggings or to dead-end connections by wedge-type sockets. Clamps, clips, and screw wedge connectors shall be permitted only on tagline and dipper trip lines.

3.15.1.6.2 Wire rope lengths. Sufficient wire rope shall be furnished for the primary and secondary drums so that with the basic boom at minimum radius shown in table VII with the hook block at ground level, a four part line, there shall be not less than three wraps of wire rope on the hoist drum. The boom hoist shall have sufficient rope to allow the tip of the boom to be lowered to rest on level ground plus three additional wraps on the drum. The primary and secondary hoist drum shall be equipped with drum lagging with a rope capacity for a 4-part hoist line and a boom length of not less than 60 feet.

Table VII. Crane function to determine rope lengths

Function	Position of boom	Position of attachment
Crane hook	At minimum radius shown in table II.	At ground level.
Clamshell	60 degrees from horizontal.	20 percent of boom length (in feet) below ground level.
Dragline	40 degrees from horizontal.	50 percent of boom length (in feet) below ground level and 30 percent of boom length (in feet) beyond boom tip.
Piledriving	At basic boom height.	Two basic boom lengths plus 50 feet below ground level.

A chart shall be provided and mounted in the cab showing the correct length and size of wire rope required for operation of all front end attachments. The chart shall be of the same material as the plate specified in 3.22.1.

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3.15.2 Secondary front-end attachments.

3.15.2.1 Boom extensions. When specified (see 6.2), boom extensions shall be furnished and shall conform to MIL-B-52203.

3.15.2.2 Boom jib. When specified, (see 6.2), the boom jib shall be furnished and shall conform to MIL-B-52207. Each boom jib when furnished shall be provided with a overhaul block and hook with safety latch actuated by stainless-steel spring.

3.15.2.3 Clamshell bucket. When specified (see 6.2), the clamshell bucket shall be furnished and shall conform to MIL-B-16028, type II, size 75. Bucket teeth shall be in accordance with MIL-T-52176. When specified, (see 6.2) side and corner teeth shall be furnished.

3.15.2.4 Tag-line. When specified (see 6.2), the tag-line shall be furnished and shall be of the reel type, with an adjustable tension mechanism, and shall control a bucket specified in 3.15.2.3, and shall be in accordance with MIL-T-52198. The tag-line shall be provided with means of attaching to the standard military boom without cutting, welding, or drilling the boom and shall be removable with common tools.

3.15.2.5 Dragline bucket. When specified (see 6.2), the dragline bucket shall be furnished and shall conform to MIL-B-16029, type II, class S, size 75. Bucket teeth shall be in accordance with MIL-T-52176.

3.15.2.6 Dragline fair-lead. When specified (see 6.2), the dragline fair-lead shall be furnished and shall conform to MIL-B-52203.

3.15.2.7 Wrecking ball. When specified (see 6.2), the wrecking ball (skull cracker) shall be furnished and shall conform to MIL-B-23276, size 1 or 2 as specified.

3.15.2.8 Pile-driver leads, boom adapter and catwalk. When specified (see 6.2), the pile-driver leads, boom adapter and catwalk shall be furnished and shall conform to MIL-L-11025. Lead sections shall consist of one top section and four lower sections for each crane-shovel.

3.15.2.9 Piledriver hammer. When specified (see 6.2), the piledriver hammer shall be furnished. The piledriver hammer shall be a drop hammer conforming to MIL-H-2514, size 4, or a diesel piledriver hammer conforming to MIL-H-21103, size as specified.

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3.15.2.10 Shovel front. When specified (see 6.2), a shovel front shall be furnished.

3.15.2.10.1 Shovel boom. The shovel boom shall be of a sturdy construction strongly reinforced and braced and shall be fabricated from structural steel shapes and plates rigidly welded to form a box section up to the shipper shaft for an inside dipper stick and to the boom tip for an outside dipper stick. Replaceable wearing shoes or other suitable fittings shall be provided where the dipper stick passes by or through the boom. Replaceable buffer rails shall be provided where the dipper strikes the boom.

3.15.2.10.2 Dipper stick. The dipper stick shall be of rigid and durable steel construction, shall resist twisting strains, and shall function at all times.

3.15.2.10.3 Shovel dipper. The shovel dipper shall be of either heavy steel plate, cast steel, or combination of both. The front lip shall be of abrasion- and wear-resistant, hard-surfaced steel. The bucket lips shall be equipped with tooth sockets, shanks, and teeth in accordance with MIL-T-52176. The dipper shall have a discharge gate (greater in area than the dipper mouth) in the bottom securely hinged to the back of the dipper with large steel hinges. The dipper front-latch opening on the wearing side shall have a replaceable wear plate where the latch bar engages. The dipper angle shall be adjustable for different digging conditions. A power trip controlled from the operator's seat and a slack takeup to prevent fouling of the trip line shall be provided. The shovel dipper capacity shall be not less than that specified in 3.9.1.2.

3.15.2.10.4 Dipper crowd and retract. The dipper crowd shall be self-adjusting at all boom angles and shall be provided with a tension adjustment for maintaining the correct tension in the chain or rope. The dipper retract and crowd shall be operated by an independent positive chain or rope mechanism.

3.15.2.10.5 Sheave. The shovel boom point sheaves shall conform to the requirements specified in 3.15.1.2.1 except that the pitch diameter shall be not less than 38 times the diameter of the rope used. Sheave shaft shall conform to the requirement specified in 3.15.1.2.2.

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3.15.2.11 Backhoe. When specified (see 6.2), a backhoe shall be furnished.

3.15.2.11.1 Boom, dipper arm, and gantry. The boom, dipper arm, and gantry shall be made of rigidly welded steel members.

3.15.2.11.2 Dipper. The backhoe dipper shall be of heavy steel plate or cast steel, shall have a cutting lip of abrasion- and wear-resistant hard-surfaced steel, and shall have the bottom reinforced and protected by the extension of the tooth sockets or wearing bars. The bucket lips shall be equipped with not less than 4 tooth shanks and teeth conforming to MIL-T-52176, size as applicable. The shank shall be pinned or bolted into the socket. Side cutters shall be replaceable cast alloy or drop forged abrasion-resistant steel, heat treated and tempered. The dipper angle shall be adjustable for different digging conditions. The bucket width measured over nominal size side cutters shall conform to table IV.

3.16 Winterization. When specified (see 6.2), the crane shall be furnished with a winterization system that will permit efficient engine heating and starting, engine coolant heating, battery heating, personnel cab heating, window defrosting, and equipment operation at any ambient temperature from minus 50° to plus 105° F. Cab doors, radiator winterfront and engine housing side panels may be removed or opened for operation above minus 25° F. Engine operation, after starting, shall not be dependent on the winterization system. Winterized cranes shall start in 5 minutes in any ambient warmer than minus 25° F. and function exactly like nonwinterized cranes without overheating or other malfunction at any temperature up to 105° F.

3.16.1 Performance (winterized crane). With the winterized crane stabilized at a temperature of not higher than minus 50° F., the winterization system shall perform as follows:

(a) Heat the engine, engine coolant, and batteries sufficiently to obtain a start within 5 minutes after 4 hours of standby 115 volt electric heat.
(b) Raise the engine operating temperature to within the ranges specified in MIL-E-11276 in 1 hour after starting.
(c) Heat the personnel cab from minus 50° to a temperature not colder than plus 35° F. in 1 hour after the engine starts, measured at 6 inches above the floor at the foot pedals, and automatically maintain the temperature of plus 35° F. or warmer.

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(d) Defrost not less than 75 percent of each front and side cab window in 1 hour, after the engine starts, and continue to maintain not less than 75 percent of each window frost free.

3.16.2 Engine heater. An electrical standby 115 volt oil pan heater shall be provided with sufficient capacity to raise the engine oil temperature in the sump to an average of plus 10° F. and assure a positive engine start after 4 hours of standby heat. The heater coil shall be installed so it is covered with oil when the oil level is at the low mark and as near the pump pickup as possible. The watt density of the heater shall not exceed 18 watts per square inch. The system shall be controlled by a 180° thermostat. The heater coil shall be covered with oil when the crane is on a 40 percent longitudinal and 30 percent side slope in all directions.

3.16.3 Coolant heater. An electrical standby 115 volt coolant heater, with a thermo-syphon system shall be provided to raise the engine coolant to plus 50° F., measured at the coolant inlet of the heater after 4 hours of standby heat. A thermostat shall be furnished and set to cut off the standby power when the coolant at the inlet of the heater reaches plus 100° F. Shutoff valves shall be installed in the lines to enable removal of the heater without draining the system.

3.16.4 Cab heater(s) and defroster(s). The cab heater(s) shall be a conventional coolant-heat-transfer type, mounted in the cab so as not to interfere with the operator(s) performing his (or their) normal functions, or obstruct entering or leaving the cab. A ducting system shall be provided to direct cab heat to the operator's feet and to defrost and maintain not less than 75 percent of each window (front and side) frost free. The controls for the heater shall be accessible to the operator and the heater thermostatically controlled to maintain the temperature requirements specified in 3.16.1. The heater shall be provided with an air intake proportioning device with connection and controls to draw any desired mixture of inside and outside air.

3.16.4.1 Coolant lines. The heater lines shall be in accordance with 3.16.11 and shall be of the size specified for the heater. Lines shall be connected so as to insure maximum circulation of the heated coolant throughout the full length of the engine cylinder block and heads. The shortest possible coolant lines shall be used and shall be supported to minimize vibration or interference with engine maintenance and operation. Flexible metallic or flexible steel-reinforced rubber hose and connections shall be used to prevent damage by line vibrations, expansion, or contraction.

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3.16.5 Battery winterization.

3.16.5.1 Battery box. The battery box shall be made of corrosion-resistant steel not less than 0.060 inch thickness. The box shall provide a minimum internal clearance of 1 inch around the perimeter and 2 inches over the top of each battery. Batteries shall be supported on nonconductive rails or multiple-posts at least 1 inch above a perforated plate over a bottom-plenum at least 1 inch deep. The box shall incorporate a gasketed cover, a drain not less than 1/2-inch ID, provision for firm, acid-resistant battery holddown and spacing, a heating-air inlet to the bottom-plenum not less than 3 inch diameter, at least 15 holes in the perforated-plate with a total cross sectional area not less than 10 square inches, and check-dampered 3-inch diameter air outlet vent. The battery supports, holddowns and spacers shall not interfere significantly with the free circulation of heating air through the spaces around the batteries. Rubber grommets or boots shall be provided to cushion and seal the opening where battery cables enter the box. The cover of the battery box shall not be used as a holddown device, shall be provided with latches and handles large enough to facilitate operation by arctic mittened hands and shall be faced with nonconductive material on the bottom side to provide a safety factor against accidental short circuiting of batteries and cables. The battery box shall be so located and installed as to allow full accessibility to the batteries for servicing and replacement and to facilitate connection of a duct from the cab-heating system. If installed in the cab, the box shall be on or near the floor, the drain shall be extended by acid and alkali resistant pipe or hose to discharge on the ground, the outlet vent shall be extended out of the cab to a downward discharging outlet. If not installed in the cab, the battery box and its cover shall be insulated externally with a 1-inch thickness of plastic foam, and the insulation shall be protected by a hard facing on the cover and on exposed sides or ends. Bottom support shall not be through the insulation unless it can be shown that its strength and rigidity are suitable for such application.

3.16.5.2 Battery heating and temperature control. The batteries shall be heated during operations by fresh air from the cab heater, delivered through a duct of at least 3-inch diameter, through a check-damper, and through a thermostatically controlled damper, the latter governed by a temperature sensing element in the battery box to maintain air-space temperature there normally between 40° and 100° F. However, a bypass or override control shall be provided, operable from the panel containing other heater controls, by which air at maximum system-temperature can be

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delivered continuously when preheating after "soakout" to extreme low temperatures. The positions of this control shall be clearly marked "BATTERY-PRE-HEAT" and "BATTERY-NORMAL-HEAT".

3.16.5.3 Battery charger. A 24-volt, 15 amp charger shall be mounted in the cab to maintain the batteries at the initial charge level during standby operations. The charger shall be equipped with a limiting device to reduce the charger output commensurate with the battery state of charge. A direct reading ammeter indicating the charger output shall be fitted to the charger case.

3.16.6 Winterized cab. The winterized cab shall be as specified in 3.12.3 and the additional requirements specified herein. The doors shall be of the swinging type and rubber sealing gaskets shall withstand temperatures of minus 50° F. to plus 105° F. Doors shall open a minimum of 90 degrees. Doors shall be equipped with positive latches of a type that will keep the doors closed when subjected to vibration. Positive hold open devices to secure doors in the open position shall also be provided. Friction or coil-spring compression devices are not acceptable. All windows, except one front window, shall be tinted safety glass conforming to MIL-G-3787, class II, except one ply shall be tinted light green color. All controls or open housings shall be covered with a boot to prevent cold air from entering the cab. A storage compartment shall be provided for the electrical 25-foot cable specified in 3.16.8.

3.16.6.1 Front window. The front window glass shall be laminated, tinted, safety-plate glass conforming to MIL-G-3787, class I, type B. The front window shall be equipped with one or more 24-volt electric window wipers. The wiper(s) shall have not less than two speeds of operation, and shall have a park position for holding the blade when the glass is stored overhead. The wipers shall clear 75 percent of the front window area when in operation and shall be capable of operation from inside the cab by handcrank.

3.16.6.2 Cab insulation. The interior of the cab shall be insulated with mastic-type insulation not less than 3/32 inch thick or polyvinyl-chloride sheet insulation not less than 1/4 inch thick. All insulation shall be nonhydroscopic, fire retardant, resistant to attack by rodents, insects and fungi, and shall maintain its form under all operating and environmental conditions specified herein.

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3.16.6.3 Weatherstripping. Weatherstripping conforming to MIL-R-14328 shall be used for sealing the doors and covers to the cab framework and shall not interfere with normal operation of the doors or covers. The type of weatherstripping and method of securing it shall prevent the entrance of blowing snow and shall reduce vibration and noise to a minimum. The weatherstripping used for sealing the bottom of the doors shall be so installed that it will not be damaged by personnel stepping on it while entering or leaving the cab.

3.16.6.4 Floor mat. The floor of that part of the cab occupied by the operator shall be fitted with a fiber glass or with polyvinylchloride shell insulation not less than 3/8 inch thick with an integral antiscuff skin surface. The floor mat shall be in sections conforming to the size of the floor plates to permit removal of floor-plate sections without distrubing adjacent areas of the mat. Floor insulation shall be bonded to floor plates.

3.16.7 Radiator winterfront. The radiator shall be furnished with a two piece removable, manually controlled metal winterfront (radiator shutter) of the sliding type. The winterfront shall not jam from blowing snow or become inoperative when covered with a sheet of ice up to and including 1/4 inch thick. The open area shall be at least 30 percent of the frontal area of the radiator core. A locking device shall be provided on each section to positively lock the winterfront in at least the following positions: Full open, half open and closed. Operations of the winterfront shall be accomplished without the use of tools or pry bars.

3.16.8 Wiring system. The electrical 115 volt heating elements for the engine sump, engine coolant and battery box shall be connected through a distribution box with a separate circuit-breaker switch for each outgoing line. This box shall be weatherproof or shall be mounted in the cab, and connected to an easily accessible, externally mounted waterproof male receptacle. A single 3-wire power cable, 25 feet long, conforming to MIL-C-3432, class II, and of adequate capacity to serve all heating elements shall be furnished. Cable terminals shall be of three-prong type and shall ground the vehicle when connected. All winterized elements and the electrical system shall be radio suppressed.

3.16.9 Air cleaner. Warm air from the engine housing may be directed through the air cleaner if necessary for more efficient engine operation.

3.16.10 Cold starting aids. The cold starting aids specified in 3.13.5 shall be installed in the cab.

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3.16.11 Hose and hose assemblies. Hose and hose assemblies shall conform to the following specifications:

Hot air duct hose.	MIL-H-8796
Coolant hose (radiator and coolant heater).	ZZ-H-428, type I or II, grade A, class 2

3.16.12 Fuels, lubricants, hydraulic fluids, and antifreeze. The crane shall be assembled, run-in, tested, and delivered with the products listed in table VIII, except that fuels and oils used during run-in and testing shall be suitable for use in the prevailing ambient temperature range.

Table VIII. Standard Military Fuels, Lubricants, Hydraulic Fluids, and Antifreeze for Winterized Cranes

Product	Specification No.	Symbol
Compound, antifreeze, arctic type.	MIL-A-11755	
Fuel oil, diesel, arctic grade.	VV-F-800	DF-A
Lubricating oil, internal-combustion engine, subzero.	MIL-L-10295	OES
Lubricating oil, gear, subzero.	MIL-L-10324	GOS
Greases, aircraft and instrument.	MIL-G-23827	GL
Greases, automotive and artillery.	MIL-G-10924	GAA
Hydraulic fluid, nonpetroleum arctic type (brake system).	MIL-H-13910	HBA
Hydraulic fluid, petroleum base.	MIL-H-5606	OHA
Lubricating oil, instrument.	MIL-L-6085	OAI

3.16.13 Fabrication. The completely winterized crane shall conform to all the requirements specified herein for nonwinterized cranes including the weight and dimensional requirements. All latches, handles, grab-handles and access doors shall be proportioned to permit grasping and manipulating by a man wearing arctic type mittens.

3.17 Lubrication. All surfaces requiring lubrication shall be provided with accessible lubrication fittings. A lubrication fitting shall be located at each area where two moving metal surfaces are in contact and under pressure. The movement shall be the result of manual operation of controls, gravity, or power-train operation.

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3.17.1 Lubricants. The crane-shovel shall operate as specified herein when lubricated with military lubricants (see 6.6). The engine and torque converter shall operate on oil conforming to MIL-L-2104. For low temperature operation, oil conforming to MIL-L-10295 may be used. Each crane-shovel shall be tagged in a conspicuous place to indicate the temperature ranges and grades of lubrication used.

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.17.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.17.4 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. Fill pipe shall be not less than 1 inch in diameter. The drain shall be fitted with a drain plug conforming to MS35844. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the crane-shovel is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. The drain plugs, the filling means, and the lubricant-level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.17.5 Lubrication service.

3.17.5.1 Grease lubrication. Bearing lubrication, including sealed bearings, shall be with greases as follows:

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(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.

(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

The crane-shovel shall be run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases may not be compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.17.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F. to 32° F. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of oils used.

3.18 Fuels and lubricants. Unless otherwise specified (see 6.2), and prior to delivery to the Government, the winterized crane-shovels shall be serviced with military fuels, lubricants, greases, hydraulic fluids, and antifreeze conforming to table VIII during manufacturing, assembly, run-in, and testing. However, ambient temperatures at the time of manufacture may dictate that lubricating oil of heavier weight than winter weight is required for run-in and operation testing.

3.19 Lifting and tiedown attachments.

3.19.1 Lifting attachments. Permanently affixed attachments that enable the crane-shovel or crane-shovel and its components complying with transportability requirements, 3.7.3, to be lifted in its normal transport position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.19.2.

3.19.2 Tiedown attachments. Permanently affixed attachments that permit fastening the crane-shovel or crane-shovel and its components complying with transportability requirements, 3.7.3, to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209.

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3.20 Railroad Humping Test. The crane-shovel or crane-shovel and its components complying with transportability requirements, 3.7.3, shall withstand a railroad humping test of 4,6, and 8 mph in one direction and 8 mph in the opposite direction without permanent deformation to any of its components.

3.21 Antirotation devices for shipping. Steel rods shall be furnished for securing the revolving superstructure to the track frame to prevent rotation during shipment. The steel rods shall be of a size to restrain a movement equal to twice the gross weight of the revolving superstructure (without boom) multiplied by the distance from the center of gravity to the center of rotation. One end of each rod shall be permanently affixed in a single position on the track frame while in use as an antirotation device and for storage. Means shall be furnished to secure the loose end of the rod to the track frame for storage.

3.22 Identification plates. Each crane-shovel and attachment shall be identified in accordance with MIL-STD-130. All plates shall be furnished by the supplier and permanently affixed by screws, bolts or rivets in a conspicuous protected location. Plates for the crane shall conform to MIL-P-514, type I, style 1, composition C, of type I grade A, class 1 material. Plates for the attachments shall conform to MIL-P-514, type I, style 4, composition C, of type I, grade A, class 1 material and shall include the following information:

(Attachment nomenclature)
FOR USE WITH
(Name of crane-shovel manufacturer(s))
(Model number or numbers of crane)
*(Range of applicable crane serial numbers)
Weight
(Center of gravity location)

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*The range of crane serial numbers will encompass the lowest machine serial number and the highest known machine serial number designed to use the attachment without modification. When an attachment is common to more than one crane model number, all applicable model numbers shall be listed.

3.22.1 Instruction plates. Instruction plates shall be furnished by the supplier and permanently affixed by screws, bolts or rivets in a conspicuous and protected location on each crane-shovel. Plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, size as applicable. The plates shall include instructions, warnings, and cautions describing the operation of the controls, any special or important procedure to be followed. Plates for transportation data shall show the silhouette of the crane-shovel in transport position indicating locations and capacity of lifting and tiedown attachments, location of center of gravity, length and size of slings required, and the recommended tiedown arrangement for the crane-shovel. The lettering shall be not less than 1/4 inch in height. The supplier shall install the electrical power warning decals furnished by the contracting officer (see 6.13).

3.22.2 Information plates. Information plates shall be furnished by the supplier and permanently affixed by screws, bolts or rivets to the crane-shovel or subassembly as required. The plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates indicating contents and capacity of fuel and oil reservoirs shall have letters not less than 1/2 inch in height and shall be attached in a visible location adjacent to the filler openings. Subassembly plates shall include the name of the part with pertinent description weight and center of gravity of each of the following:

(a) Crane-shovel boom.
(b) Cab.
(c) Counterweight.
(d) Crawler base.
(e) Revolving superstructure.
(f) Swing circle or roller path.

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3.23 Treatment and painting. All parts of the crane normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color 24087 as specified in FED. STD. No. 595.

3.24 Stenciling and vehicle weight classification.

3.24.1 Stenciling. Lifting and tiedown attachments shall be identified with 1 inch letters indicating "sling point" or "tiedown". Unless otherwise specified herein, stenciling shall conform to MIL-T-704. Registration numbers shall be marked on the crane-shovel in accordance with MIL-STD-642. The color shall be white.

3.24.2 Crane weight classification. Unless otherwise specified (see 6.2), the supplier shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the crane as specified (see 6.10).

3.25 Operating accessories. Two 20-foot-long chains conforming to RR-C-271, type I, grade C, class 2, of 3/4-inch stock shall be furnished for each crane-shovel. Each chain shall be provided with a 6-inch-inside-diameter ring on one end and a grab hook on the other end. Provision shall be made for carrying the chain on the crane-shovel.

3.26 Items to be furnished with crane-shovel. No shipment of crane-shovels shall be made unless repair parts, operating tools, and special maintenance tools and accessories are included with the shipment or unless approval for shipment of the crane-shovels without such items has been received from the contracting officer.

3.27 Government-furnished property. When specified (see 6.2), a rifle bracket assembly FSN 2540-763-7348, shall be mounted within the cab in a location which is accessible to the operator. The kit will be furnished by the Government (see 6.7), but all fasteners and fittings used shall be furnished by the supplier.

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3.28 Toolbox. A toolbox made of steel not less than 0.071 inch thick shall be attached to the crane-shovel. The toolbox shall be sufficiently large to contain all the tools required for the routine maintenance and operation of the crane-shovel. The box lid shall be secured to one edge of a side by a continuous hinge conforming to MS35829-7C extending the full length of the box edge. The toolbox hinged cover shall be held in the closed position by two tension-type adjustable hook latches with strikers not less than 3 inches in length. A hasp shall be suitably located to secure the lid to the box and accommodate a lock conforming to MS35647. The toolbox shall be securely mounted in an accessible, protected location permitting the lid to be opened through an arc of not less than 90 degrees. The lid shall be held fully open by the weight of the lid or a suitable fastener. Drainage of the toolbox shall be provided as mounted.

3.29 Government-loaned property. When specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8):

Item No.	Description	Identification (FSN)	Quantity
1.	Lead Section, Top, Pile Driver: 15 Ft Lg.	3815-983-8029	1
2.	Lead Section, Lower, Pile Driver: 10 Ft. Lg.	3815-221-2215	4
3.	Boom, Crane, 30 Ft: Complete Boom, Both Sections	3815-541-9839	1
4.	Boom Extension, Middle, Crane: 10-Ft Extension	3815-540-4756	3
5.	Boom Jib, Crane: 15-Ft Jib	3815-542-4839	1
6.	Adapters, Pile-Driver Lead	3815-554-0083	1 set
7.	Catwalk, Piledriver: Telescoping; 3 Sect; 8 to 23 Ft.	3815-190-3308	1
8.	Fairlead, Roller and Sheave: for 12-1/2-Ton Crane-Shovel, Crawler-Mounted and 20-Ton Crane-Shovel, Truck-Mounted.	3815-554-0126	1
9.	Bucket, Dragline: 3/4 Cu Yd.	3815-186-3571	1
10.	Bucket, Clamshell: 3/4 Cu Yd.	3815-186-3556	1
11.	Tagline, Crane and Crane-Shovel: 3/4 to 1 Cu Yd Bucket	3815-249-8387	1

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3.30 Servicing and adjustment. The supplier shall service and adjust each crane-shovel for operational use including at least the following: Adjustment of brakes and mechanical components of the engines, transmission, drive train, servicing of the cooling system with a 50-50 solution, by volume, of ethylene-glycol-type antifreeze conforming to MIL-A-46153, and water and complete lubrication as specified in 3.17 unless otherwise specified in section 5.

3.31 Workmanship. All parts, components, and assemblies of the crane-shovel including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required.

3.31.1 Steel fabrication. Steel used in the fabrication of the equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.31.2 Bolted connections. Bolt holes shall be accurately formed or drilled and burrs and sharp edges shall be removed. All bolts, nuts, and screws shall be tight. Adjoining surfaces shall be flat, and in full contact with each other. Lockwashers shall be provided for all bolts, nuts, and screws.

3.31.3 Riveted connections. Rivet holes shall be accurately formed or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members and shall be in accordance with SAE J492.

3.31.4 Welders and welding.

3.31.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

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Standard Qualification Procedure of the American Welding Society. Welding Qualifications of the ASME.

3.31.4.2 Welding. All welding shall be in accordance with the accepted practice of industry in accordance with the American Welding Society B3.0, as specified on the drawings and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses.

3.31.4.3 Mechanical properties. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification.

3.31.5 Machine work. Tolerance and gages for metal fits shall conform to the limitations specified herein.

3.31.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or any other defect which might render the casting unsound for its intended use. Rework shall be limited to repairs which do not reduce the strength or affect function.

3.31.7 Preparation of liquid and air systems. The hydraulic and water systems shall all be purged to remove any foreign material in the lines or components after the assembly of the crane-shovel is complete.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination.

4.3.1.1 Pretest examination. The preproduction crane-shovel shall be subjected to the examination marked "X" under column 1 of table IX. When winterized crane-shovels are furnished, the winterized model shall also be subjected to the examination marked "X" under column 1, of table X. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the crane-shovel shall be subjected to the examinations marked "X" in column 2 of table IX and 4.6.2.29. When winterized crane-shovels are furnished, the winterized model shall also be subjected to the examinations marked "X" under column 2 of table X. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The nonwinterized preproduction crane-shovel shall be subject to test marked "X" in column 2 of table XI. Winterized preproduction crane-shovel shall be subject to tests marked "X" in column 2 of tables XI and the tests in table XII. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production crane-shovel(s) will be selected at random by the Government from the crane-shovels being produced by production tooling and will be examined for the defects marked "X" in columns 3 and 4 of tables IX and X, as applicable and subjected to the tests marked "X" in column 2 of table XI to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production crane-shovel shall not exclude the remaining crane-shovels from the quality conformance inspection and acceptance provisions specified

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in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production crane-shovel to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production crane-shovel(s) and shall be cause for refusal by the Government to continue acceptance of production crane-shovels until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on crane-shovels previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all crane-shovels accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each crane-shovel shall be examined for the defects marked "X" in column 5 of tables IX and X (for winterized crane-shovel). Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crane-shovel shall be subjected to the tests marked "X" in column 3 of table XI. Failure of any test shall be cause for rejection.

4.6 Inspection procedure. Prior to examination or tests, the crane-shovel shall be completely lubricated with military oils and military greases as specified in 3.17. Oils shall be those designated for use in the ambient temperatures at which the tests are being performed.

4.6.1 Examination.

4.6.1.1 Nonwinterized crane-shovel. Examination shall be in accordance with table IX.

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Table IX. Examination Schedule (Nonwinterized)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	X	X	X	-	101. Any component missing, damaged, or not as specified.	3.1
X	-	-	-	-	102. Weights or dimensions not as specified.	3.1.1
X	X	X	X	-	103. Compatibility not as specified.	3.1.2
X	-	-	-	-	104. Material identification not as specified.	3.2.3
X	-	-	-	X	105. Material not as specified.	3.4
X	-	-	-	X	106. Interchangeability of parts not as specified.	3.4.1
X	-	-	-	-	107. Threaded parts not as specified.	3.4.8
X	X	X	X	-	108. Safety characteristics not as specified.	3.5.1
X	X	X	X	-	109. Human factors and sound level requirements not as specified.	3.5.2, 3.5.3
X	X	X	X	-	110. Maintainability not as specified.	3.6
X	X	X	-	X	111. Filling, draining or checking provisions not as specified.	3.6.2

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	-	-	-	112. Transportability not as specified.	3.7
X	X	X	X	-	113. Environmental and/or use capabilities not as specified.	3.8
X	-	X	-	-	114. Crane capacity not as specified.	3.9.1.1
X	-	X	-	-	115. Shovel operating range not as specified.	3.9.1.2
X	-	X	-	-	116. Backhoe working range not as specified.	3.9.1.3
X	-	X	-	-	117. Clamshell or dragline not compatible as specified.	3.9.1.4
X	X	X	-	X	118. Crane capacity plate not as specified.	3.9.1.5
X	-	X	-	X	119. Maneuverability not as specified.	3.9.2.1
X	-	-	-	-	120. Ground pressure not as specified.	3.9.2.2
X	-	-	-	-	121. Stability not as specified.	3.10
X	-	-	-	-	122. Maximum unit stress exceeding limits as specified.	3.11
X	-	-	-	-	123. Superstructure not as specified.	3.12

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	-	-	-	124. Platform, catwalks, footholds or handholds not as specified.	3.12
X	-	-	-	-	125. Piledriver lead catwalk clips not as specified.	3.12
X	-	-	-	-	126. Counterweight not as specified.	3.12
X	-	-	-	X	127. Functional requirements not as specified.	3.12.1
X	-	-	-	-	128. Forward deck machinery protection not as specified.	3.12.2
X	-	-	-	-	129. Cab not as specified.	3.12.3
X	-	-	-	-	130. Cab heating not as specified.	3.12.3.1
X	X	X	-	-	131. Superstructure cab disassembly not as specified.	3.12.4
X	X	X	-	-	132. A-Frame not as specified.	3.12.5
X	-	-	-	-	133. Power load-lowering device not as specified.	3.12.6
X	-	-	-	-	134. Crane-boom hoist not as specified.	3.12.7

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	-	-	-	135. Drums not as specified.	3.12.8
X	-	-	-	-	136. Hoist drum brakes not as specified.	3.12.8.1
X	-	-	-	-	137. Hydraulic components not as specified.	3.12.8.2
X	X	-	-	-	138. Drum clutches not as specified.	3.12.8.3
X	X	-	-	-	139. Rotating frame not as specified.	3.12.9
X	X	-	-	-	140. Swing-gear or lock or brake mechanism not as specified.	3.12.10
X	-	-	-	-	141. Power transmission clutch not as specified.	3.12.11.1
X	-	-	-	-	142. Torque converter not as specified.	3.12.11.2
X	X	-	-	-	143. Power transmission chains not as specified.	3.12.11.3
X	X	X	X	X	144. Electrical system not as specified or does not function.	3.12.12

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	X	-	X	X	145. Fungus and moisture resistance not as specified.	3.12.12.1
X	X	-	-	-	146. Wiring not as specified.	3.12.12.3
X	X	-	-	-	147. Floodlights not as specified.	3.12.12.4.1 thru 3.12.12.4.1.2
X	X	-	-	-	148. Interior cab lights not as specified.	3.12.12.4.2
X	X	-	-	-	149. Warning horn not as specified.	3.12.12.5
X	-	X	-	-	150. Trouble lamp not as specified.	3.12.12.6
X	X	-	-	-	151. Instrument panel not as specified.	3.12.12.7
X	X	-	X	-	152. Operator controls not as specified.	3.12.13
X	X	-	X	-	153. Crane controls not as specified.	3.12.13.1
X	X	-	X	-	154. Engine hand controls not as specified.	3.12.13.2
X	X	X	X	-	155. Accumulator not as specified.	3.12.13.3
X	X	-	-	-	156. Engine not as specified.	3.13

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	X	-	-	-	157. Power and speed rating not as specified.	3.13.1
X	X	X	X	-	158. Lubrication system not as specified.	3.13.3
X	X	-	-	-	159. Fuel system not as specified.	3.13.4
X	X	-	-	-	160. Cold starting aids not as specified.	3.13.5
X	X	-	-	-	161. Coolant system not as specified.	3.13.6
X	X	-	-	-	162. Air induction system not as specified.	3.13.7
X	X	-	-	-	163. Engine exhaust system not as specified.	3.13.8
X	-	-	-	-	164. Governor not as specified.	3.13.9
X	-	-	-	-	165. Starting system not as specified.	3.13.10
X	-	-	-	-	166. Charging system not as specified.	3.13.11
X	-	-	-	-	167. Batteries not as specified.	3.13.12
X	-	-	-	-	168. Battery container not as specified.	3.13.12.1

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	X	-	-	-	169. Emergency starting or charging receptacle or cable not as specified.	3.13.13
X	-	-	-	-	170. Crawler base not as specified.	3.14.1
X	-	-	-	-	171. Track assembly not as specified.	3.14.2
X	-	-	-	-	172. Side frames not as specified.	3.14.3
X	-	-	-	-	173. Crawler drive not as specified.	3.14.4
X	-	-	-	-	174. Axle not as specified.	3.14.5
X	X	X	-	X	175. Front-end attachment not as specified and components missing.	3.15
X	X	-	-	-	176. Crane boom not as specified.	3.15.1.1
X	X	-	-	-	177. Pendant equilizer assembly not as specified.	3.15.1.2 thru 3.15.1.2.3
X	X	-	-	-	178. Boom safety stop not as specified.	3.15.1.3

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	-	-	X	179. **Boom angle indicator, chart, or overload indicating system not as specified.**	3.15.1.4, 3.15.1.4.1
X	X	-	-	-	180. Crane hook block not as specified.	3.15.1.5
X	X	X	-	-	181. Rope not as specified.	3.15.1.6
X	-	-	-	-	182. Rope sockets not as specified.	3.15.1.6.1
X	-	-	-	X	183. Rope length or rope length chart not as specified.	3.15.1.6.2
X	-	X	-	X	184. Secondary front-end attachment not as specified or components missing.	3.15.2
X	-	-	-	-	185. Boom extension not as specified.	3.15.2.1
X	-	-	-	-	186. Boom jibs not as specified.	3.15.2.2
X	-	-	-	-	187. Clamshell buckets not as specified.	3.15.2.3
X	-	-	-	-	188. Tag-lines not as specified.	3.15.2.4
X	-	-	-	-	189. Dragline bucket not as specified.	3.15.2.5

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	-	-	-	190. Dragline fairlead not as specified.	3.15.2.6
X	-	-	-	-	191. Wrecking ball not as specified.	3.15.2.7
X	X	-	-	-	192. Piledriver leads, boom adapters or catwalks not as specified.	3.15.2.8
X	-	-	-	-	193. Piledrivers not as specified.	3.15.2.9
X	X	X	-	X	194. Shovel fronts not as specified and components missing.	3.15.2.10 thru 3.15.2.10.5
X	X	X	X	X	195. Backhoes not as specified or components missing.	3.15.2.11
X	X	X	-	X	196. Lubrication, lubricants or components not as specified.	3.17 thru 3.17.5.2
X	X	X	X	-	197. Lifting attachments not as specified.	3.19.1
X	X	X	-	-	198. Tiedown attachments not as specified.	3.19.2
-	X	-	X	X	199. Antirotation devices not as specified.	3.21

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Table IX. Examination Schedule (Nonwinterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	X	-	-	X	200. Identification, instruction or information plates not as specified.	3.22, 3.22.1, 3.22.2
X	X	X	-	X	201. Treatment or painting not as specified.	3.23
X	X	X	-	X	202. Stenciling not as specified.	3.24
X	-	X	X	X	203. Operating accessories not as specified.	3.25
X	-	X	-	X	204. Item to be furnished with crane-shovel not as specified.	3.26
X	-	X	-	-	205. Rifle mounting kit not as specified.	3.27
X	X	X	-	-	206. Toolbox not as specified.	3.28
X	X	X	X	X	207. Workmanship not as specified.	3.31 thru 3.31.7
X	-	X	-	X	208. Servicing or adjustment not as specified.	3.30

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4.6.1.2 Winterized crane-shovels. Examination shall be in accordance with table X.

Table X. Examination Schedule (Winterized)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	X	-	X	209. Insulation in both cabs and engine housings is missing. Thickness, including insulation under the floormats not as specified.	3.16.6.2
X	-	X	-	-	210. Defrosters not as specified.	3.16.4, 3.16.1
X	-	X	-	-	211. Battery boxes not as specified or not of corrosion-resistant steel. Battery boxes not provided with drainage.	3.16.5.1
X	-	X	-	-	212. Winterized components not in protected locations or not protected.	3.16
-	X	-	X	-	213. Thermal damage (caused by heating media) to batteries.	3.16.5.2
X	-	X	-	-	214. Winterized components not as specified.	3.16

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Table X. Examination Schedule (Winterized) - (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	X	-	-	215. Positioning of winterized components not as specified.	3.16.4
X	-	X	-	-	216. Cab not weatherstripped as specified.	3.16.6.3
X	-	X	-	-	217. Floormat not as specified.	3.16.6.4
X	-	X	-	-	218. Boots and grommets not as specified.	3.16.6
X	-	X	-	-	219. Window wiper not as specified.	3.16.6.1
X	-	X	-	-	220. Cab heating system not as specified.	3.16.4
X	-	X	-	-	221. Heater mounting or enclosure not as specified.	3.16.4
X	-	X	-	-	222. Winterfront not as specified.	3.16.7
X	-	X	-	-	223. Engine heater not as specified.	3.16.2
X	-	X	-	-	224. Coolant heater not as specified.	3.16.3
X	-	X	-	-	225. Battery heater not as specified.	3.16.5.2
X	-	X	-	-	226. Battery winterization not as specified.	3.16.5

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Table X. Examination Schedule (Winterized) (Cont'd)

Preproduction		Initial production		Quality conformance	Defect	Requirement Paragraph
Pretest	Post test	Pretest	Post test			
1	2	3	4	5	6	7
X	-	X	-	-	227. Thermostatic controls not furnished or do not operate as specified.	3.16.2, 3.16.3, 3.16.4, 3.16.5.2
X	-	X	-	X	228. Wiring system or components not as specified.	3.16.8

Table XI. Test Schedule (Nonwinterized)

Prepro.	Init. Prod.	Qual. Conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	X	Test Conditions	4.6.2.1	3.9.3., 3.17
X	-	-	Engine	4.6.2.2	3.2.1.1
X	X	-	Accumulator	4.6.2.3	3.12.13.3
X	-	-	Disassembly and assembly for transport	4.6.2.4	3.7
X	-	-	Ease of maintenance	4.6.2.5	3.6
X	-	-	Emergency starting	4.6.2.6.1	3.12.12
X	-	X	Lights and horn	4.6.2.6.2	3.12.12, 3.12.12.5
X	-	-	Lubrication	4.6.2.7	3.6, 3.17
X	-	-	Lifting, tiedown and rail humping	4.6.2.8 thru 4.6.2.8.3.2	3.19, 3.20
X	-	-	Antirotation	4.6.2.9(a)	3.21
X	-	-	Swing lock	4.6.2.9(b)	3.12.10
X	-	-	Swing brake	4.6.2.9(c)	3.12.10

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Table XI. Test Schedule (Nonwinterized) - (Cont'd)

Pre-pro.	Init. Prod.	Qual. Conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Maneuverability	4.6.2.10	3.9.2.1
X	-	-	Grade	4.6.2.10.1	3.9.2.1(a)
X	-	-	Slope	4.6.2.10.2	3.9.2.1(b)
X	-	-	Brake	4.6.2.10.3	3.9.2.1(a)
X	-	-	Steering	4.6.2.10.4	3.9.2.1
X	X	-	Propelling test	4.6.2.10.5	3.9.2.1(c)
X	X	-	Fording	4.6.2.10.6	3.9.2.1(d)
X	-	-	Ground bearing pressure	4.6.2.11	3.9.2.2
X	-	-	Backward stability	4.6.2.12	3.10
X	-	-	Maximum working stress	4.6.2.13	3.11
X	-	-	Lifting capacity	4.6.2.14	3.9.1.1
X	X	-	Craning	4.6.2.15	3.6.2
X	-	-	Crane capacity plate determination	4.6.2.16	3.9.1.5
X	-	-	Overload indicating system	4.6.2.16.1	3.15.1.4.1
X	-	-	Rope strength	4.6.2.17	3.15.1.6
X	-	-	Power and speed test	4.6.2.18	3.8.3, 3.12.1, 3.15.1.6.2
X	-	-	Swing	4.6.2.19	3.12.1
X	-	-	Power lowering	4.6.2.20	3.12.6
X	-	-	Clutch capacity	4.6.2.21	3.12.6
X	-	-	Backing down	4.6.2.21.1	3.12.6
X	-	-	Clutch control	4.6.2.21.2	3.12.8.3, 3.12.13.3
X	-	-	Boom hoist cable angle	4.6.2.22	3.12.5
X	-	-	Boom stop and pile lead	4.6.2.23	3.15.1.3
X	-	-	Electromagnetic interference	4.6.2.24	3.12.12.2
X	-	-	Counterweight removal	4.6.2.25	3.12
X	X	-	Backhoe	4.6.2.26.1	3.9.1.3

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Table XI. Test Schedule (Nonwinterized) - (Cont'd)

Pre-pro.	Init. Prod.	Qual. Conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	-	Shovel front	4.6.2.26.2	3.9.1.2
X	X	-	Clamshell and dragline	4.6.2.27	3.9.1.4
X	-	-	High temperature operation (nonwinterized unit only)	4.6.2.28.1.1	3.8.1
X	-	-	Low temperature operation	4.6.2.28.1.2	3.8.1
X	-	-	High temperature storage	4.6.2.28.2.1	3.8.2
X	-	-	Low temperature storage	4.6.2.28.2.2	3.8.2
X	-	-	Rain	4.6.2.28.3	3.12
X	-	-	Post test disassembly	4.6.2.29	3.12
-	-	X	Brake and clutch	4.6.2.30.1	3.12
-	-	X	Swing	4.6.2.30.2	3.12
-	-	X	Hoisting	4.6.2.30.3	3.12
-	-	X	Boom hoisting	4.6.2.30.4	3.12
-	-	X	Boom stop	4.6.2.30.5	3.15.1.3
-	-	X	Off road	4.6.2.30.6	3.9.2
-	-	X	Overload system	4.6.2.30.7	3.15.1.4.1
X	-	-	Sound level	4.6.2.31	3.5.3
X	-	-	Cooling	4.6.2.32	3.13.6
-	X	-	Mean-time-between failure	4.6.2.33	3.8.5
-	X	-	Maintenance ratio	4.6.2.34.1	3.6.1
-	X	-	Maintenance package	4.6.2.34.2	3.6, 3.17
X	-	-	Torque converter stall (when furnished)	4.6.2.35	3.12.11.2

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Table XII. Test Schedule (Winterized)

Sequence	Prepro. tests	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
1	X	High temperature operation	4.6.2.28.4.1	3.16
2	X	Cold starting and running	4.6.2.28.4.2	3.16

4.6.2 Tests. Tests shall be in accordance with table XI for non-winterized models. In addition, winterized models shall also be subjected to tests specified in table XII. An "X" in a column indicates the tests that shall be conducted.

4.6.2.1 Test conditions. Prior to test, the crane-shovel shall be lubricated with oils and greases specified in 3.17.5. Oils shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The crane-shovel shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test.

(a) The front-end equipment required for the tests specified herein but not included as Government-loaned equipment (see 3.29) shall be furnished by the supplier at no cost to the Government. The equipment furnished by the supplier for the test shall conform to the applicable military specification for the equipment.

(b) The test area shall be examined to determine that:

(1) There are no obstructions which may cause interference to raising the boom to its maximum height in any position or rotating the boom horizontally through 360 degrees.

(2) Grade does not exceed 1 percent in any direction.

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(3) That the test is to be performed on a hard surface.

(c) The crane shall be checked prior to and during the tests to determine the following:

(1) Insure that coolant and lubrication systems are full.

(2) Check that fuel tanks are maintained between full and half-full level throughout these tests. Check fuel gages for accuracy at empty, one-half full, and full levels.

(3) Adjust clutches and brakes in accordance with the manufacturer's instruction manual. Inspection shall be made to insure that there is no dragging of the clutches or brakes with the controls in the disengaged position.

(4) Examine all cable anchors, sheaves, revolving base rollers, drive chains, rotating gears, rotating drums and ropes to insure that there is no rubbing or interference. This examination shall be made at all operating positions of the front-end attachments and shall include the boom raised to maximum height.

(5) All lights shall be turned on during all operational tests.

(d) The component tests 4.6.2.2 and 4.6.2.3 shall be conducted prior to the preproduction model tests of the complete crane-shovel.

4.6.2.2 Engine. Prior to installation of the engine in the preproduction model(s), the engines shall be tested in accordance with MIL-STD-1400, test method 2000. Nonconformance to 3.13 shall constitute failure of the test. Upon successful completion of this test and prior to installation of the engine in the preproduction model(s), the fuel injection pumps, governors, and turbocharger, if furnished, shall be sealed by a Government representative at the settings established by this test. Seals for each engine shall also be placed by a Government representative to prevent separation of the crankcase from the engine block and the rocker-arm cover from the engine head. Any changes made to these settings or removal or alteration of the seals by other than the Government inspector or without his expressed consent will invalidate all tests previously performed on these engines.

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4.6.2.3 Accumulator.

(1) Pretest accumulator charge pressure. When a power hydraulic control system is furnished for the crane-shovel (see 3.12.13.3), the accumulator charge pressure shall be measured prior to performing any tests.

(2) Post test accumulator charge pressure. After completion of all tests, when a power hydraulic control system for the crane-shovel is furnished, the accumulator charge pressure shall be measured. An accumulator charge pressure less than that measured in (1) above shall constitute failure of the test.

4.6.2.4 Disassembly and assembly for transport.

(a) Disassemble the crane-shovel (using a crane) into components for air transport as specified in 3.7.2. Pass the components through a rigid mock-up ramp for aircraft loading as shown on figure 1. Then reassemble the crane-shovel. Failure to disassemble in 6 hours and reassemble in 6 hours or inability to pass the components through the mock-up ramp with less than the specified clearance shall constitute failure of this test.

(b) Disassemble the crane-shovel into components, for rail transport and load on flat car as specified in 3.7.3. Pass the components through a rigid mock-up in accordance with Berne International Clearance Diagram (see figure 2). Then reassemble the crane-shovel. Failure to disassemble in 6 hours and reassemble in 6 hours or inability to pass the components through the mock-up shall constitute failure of this test.

4.6.2.5 Ease of maintenance. Maintenance operations listed in 3.6 shall be accomplished on the crane-shovel. Failure to accomplish the operation in the time specified with the tools specified shall constitute failure of this test. The use of a hoist with its operator will be permitted during engine removal and replacement.

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4.6.2.6 Electrical.

4.6.2.6.1 Emergency starting. Disconnect the crane batteries and start the engine using the emergency charging receptacle and 20 foot long cable, as specified in 3.12.12 connected to a separate source of power. Failure of the use of the cable connection to start the engine within 1 minute shall constitute failure of this test.

4.6.2.6.2 Lights and horn. Operate each of the switches controlling all lights and the horn button for not less than five times. Measure the light intensity of the instrument panel light. Failure of the switches to operate, lights to burn clearly, horn to function properly, light intensity of instrument panel light less than that specified shall constitute failure of this test.

4.6.2.7 Lubrication test. Lubricate the crane as recommended by the manufacturer. Inability to conform to the following shall constitute failure of this test:

(a) That the seals retain lubricant.
(b) That all fittings are accessible without removal of parts or components of the crane.
(c) That fittings comply with 3.17.2 and 3.17.3.
(d) That fittings are protected from damage due to operation or movement of the crane.
(e) That provision has been made for lubrication of gears, chains, and sprockets.

4.6.2.8 Lifting and tiedown attachments, and rail humping.

4.6.2.8.1 Lifting attachments.

4.6.2.8.1.1 Test procedures.

(a) Lift the crane-shovel or crane-shovel and its components complying with transportability requirements, 3.7.3, and hold in suspension in its normal transport position using slings that converge not more than 24 feet above the lowest extremity of the crane-shovel or crane-shovel and its components. Determine the forward angles of application for each attachment. Measure the clearance between each sling and the crane-shovel or crane-shovel and its components. Measure eye openings and clearance dimensions of the attachments.

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(b). Restrain the crane-shovel or crane-shovel and its components and subject each attachment to a force of 2-1/2 times the load it carried when initially suspended. This force will be applied in the direction as determined in (a) above. An alternate method of test is to lift the crane-shovel or crane-shovel and its components as in (a) above and add weights to the main frame until the force is 2-1/2 times the initial force on each attachment. Hold the crane-shovel or crane-shovel and its components under either of the above loading conditions for at least 90 seconds.

4.6.2.8.1.2 Failure criteria. Weld failure or permanent deformation of the crane-shovel or crane-shovel and its components at points other than the lifting attachments, or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.6.2.8.2 Tiedown attachments.

4.6.2.8.2.1 Test procedure. With the crane-shovel or crane-shovel and its components anchored by means other than the attachment being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedowns.

4.6.2.8.2.2 Failure criteria. Breakage or failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.6.2.8.3 Railroad humping test.

4.6.2.8.3.1 Test procedure. Anchor the crane-shovel or crane-shovel and its components on a domestic service standard railway flatcar (blocking is permitted). Hump the car with the crane-shovel or crane-shovel and its components tied down, at speeds of not less than 4, 6, and 8 mph against five empty boxcars with their brakes set. The crane-shovel or crane-shovel and its components shall face in the direction of the five cars. Repeat the procedure at not less than 8 mph with the rear of the crane-shovel or crane-shovel and its components in the direction of the five cars. The crane-shovel or crane-shovel and its components may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the crane-shovel or crane-shovel and its components in the proper direction on the flatcar. Once the test has begun, there will be no readjustment of load, nor any reconditioning of the blocks, checking material or tiedowns except as necessary to reverse the direction of the crane-shovel or crane-shovel and its components. After each impact, the crane-shovel

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or crane-shovel and its components shall be inspected for permanent deformation to its components or other evidence of possible failure. After completion of the railroad impact test an operational checkout of the crane-shovel or crane-shovel and its components will be conducted.

4.6.2.8.3.2 Failure criteria. Weld failure or permanent deformation to the components of the crane-shovel or crane-shovel and its components or damage to any tiedown cable or stopping of a test due to failure or damage to the crane-shovel or crane-shovel and its components or failure of the crane-shovel or crane-shovel and its components to operate after the test shall constitute failure of the test.

4.6.2.9 Antirotation test.

(a) Install the antirotation rods and tie the crane down to prevent tipping. Release the swing brake and lock of the crane. Reeve a line to a point on one end of the revolving superstructure to exert a moment in a horizontal plane both clockwise and counterclockwise. The moment shall be not less than twice the gross weight of the superstructure (without boom) times the distance from the center of gravity to the center of rotation.

(b) Repeat the procedure with the swing lock engaged and the antirotation rods and swing brake released with a moment equal to one-half the moment in (a) above.

(c) Repeat the procedure with a moment of not less than 0.30 of the torque used above with the swing brake set and the antirotation rods and swing lock released.

Breakage or distortion of the rods, brackets, attaching members, or welds or inability of either the lock or brake independently to hold the superstructure in place shall constitute failure of this test.

4.6.2.10 Maneuverability test. With fuel tank not less than 3/4 full perform the following.

4.6.2.10.1 Grade. The crane-shovel shall ascend and descend a grade of not less than 30 percent and 50 feet in length with each attachment

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listed in column 1 mounted on the crane-shovel, with the counterweight located as listed in column 2, and with the crawler track travel as shown in column 3 in table XIII.

Table XIII. Grade test.

Attachments	Counterweight location	Crawler track travel
Col. 1	Col. 2	Col. 3
Boom, 30 feet with block	Forward	Forward
Backhoe	Back	Forward
Shovel	Back	Forward

Inability of the crane-shovel to negotiate the grade, permanent deformation, fuel leakage or to complete the test shall constitute failure of this test.

4.6.2.10.2 Slope. Self propel the crane-shovel, with operator on the high side, on a side slope of not less than 15 percent grade and not less than 100 feet in length. Repeat the above with operator on the low side. Inability of the crane-shovel to negotiate the side slope, permanent deformation, fuel leakage or inability to complete the test shall constitute failure of this test.

4.6.2.10.3 Brake. The crane-shovel shall be equipped with the front end attachment to provide the maximum crane-shovel weight. With the crane-shovel track in forward travel, ascend the grade specified in 4.6.2.10.1, stopping three times on the grade and engage and lock the track locking mechanism at each stop. Descend the above grade, stopping three times on the grade and engage and lock the track locking mechanism at each stop. Inability of the track locking mechanism to hold the tracks stationary or permanent deformation shall constitute failure of this test.

4.6.2.10.4 Steering. Steer the crane-shovel to the right and left to prove the steering capability. Inability to steer shall constitute failure of this test.

4.6.2.10.5 Propelling test. Self propel the crane-shovel with 30 foot boom and hook block for 40 miles at an average speed of not less than 0.75 miles per hour. Terrain covered shall consist of 20 miles of unimproved road and 20 miles of off road terrain. Off road terrain shall include not less than a total of 1 mile of 10 percent grade. Inability to complete the test, permanent deformation, breakage or malfunction shall constitute failure of this test.

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4.6.2.10.6 Fording. Water fording capabilities shall be ascertained as follows for the preproduction model: (a) Bring crane to operating temperature. (b) Operate a minimum of 2 hours with not less than 12-inch deep water. (c) Sample oil from engine and all drive components. (d) Repeat above four times except the crane shall be stopped a minimum of four times. The initial production test items shall be subjected only to (a) and (b) above. Inability to complete or nonconformance to 3.9.2.1(d) shall constitute failure of this test.

4.6.2.11 Ground bearing pressure test. The ground bearing pressure is derived by dividing the total working weight with complete front-end equipment, but not including load by the bearing area in square inches. The weight shall be determined by platform scales, by calibrated hydraulic, or electronic devices and the bearing area used shall be the center-to-center distance between crawler end sprockets multiplied by twice the width of the track shoe. Ground bearing pressure over 11 psi shall constitute failure of this test.

4.6.2.12 Backward stability test. With the boom of the length specified in table I at its minimum working radius in the least stable direction, the location of the center of gravity shall be determined as follows for conformance to 3.10:

$$X = \frac{Rd}{W}$$

Where: X = Distance between centerline of crawler track roller opposite lifting side of crane to center of gravity.

R = Weight measured when crawler track opposite lifting side of crane-shovel is on firm ground and the crawler track on lifting side of crane-shovel is on a platform scale.

d = Distance between centerlines or on center of outer flange of the crawler track rollers.

W = Weight measured when entire crane-shovel is on platform scales.

Failure to comply with the requirement in 3.10 shall constitute failure of this test.

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4.6.2.13 Maximum working stress. The maximum working stress test shall be conducted as follows:

(a) Preparation of crane.

(1) One-half-full to full fuel tanks.
(2) Full coolant and lubricant.
(3) Installation of crane boom, shovel front and backhoe.
(4) All exterior surfaces cleaned of grease, oil, and dirt.

(b) Installation of strain gages. Strain gages shall be installed in accordance with section 54, Resistance Strain Gage Testing, Non-Destructive Testing Handbook. Where directions of principal stress are questionable, two or more gages or strain-gage rosettes may be employed. During test operation or visual examination, any points which show visual evidence of weakness due to design or paint checking should be gage tested in addition to those points listed in 4.6.2.13(d). Particular note shall be made of residual stresses in the structural members due to dead loading. These stresses shall be relieved prior to zeroing the strain-recording instrument (i.e., resting the crane boom on a support with cable slacked).

(c) Instrumentation operation. Operation of strain-gage recording or indicating apparatus shall be in accordance with the instrument manufacturer's recommended procedure. Standard commercial instruments specifically designed for resistance strain gages shall be considered acceptable provided the manufacturer guarantees an accuracy to within plus or minus 10 microinches at a scale reading of 600 microinches.

(d) Location of strain gages. Strain gages shall be located approximately as listed:

Superstructure:

S-1 *Front hook roller bracket.
S-2 *Rear hook roller bracket.

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S-3	Hook roller path - Top side, outer edge, radially.
S-4	*Hook roller path - Top side, outer edge, tangential.
S-5	Boom foot brackets.
S-6	A-frame compression members - Top midsection.
S-7	A-frame compression members - Side midsection.
S-8	A-frame tension members - Rear midsection.
S-9	A-frame tension members - Side midsection.
S-10	A-frame spreaders - Top midsection.
S-11	A-frame spreaders - Front midsection.
S-12	A-frame head shaft - Top midsection.
S-13	A-frame head shaft - Front midsection.
S-14	Other critical points on A-frame.
S-15	*Drum mountings - At points of critical stress.
S-16	*Rotating base - At points of critical stress (i.e., sills).
S-17	*Pendant equalizer - At points of critical stress.

Note any cut-away sections or direction changes in structural members which may be points of stress concentration.

Shovel front:

SF-1	Boom head shaft - Top side.
SF-2	Boom head shaft - Front side.
SF-3	*Dipper stick - Top side to rear of bucket.
SF-4	Other critical points of shovel-front attachment.

Backhoe:

H-1	Sub A-frame - Midsection main compression members.
H-2	Sub A-frame - Midsection spreaders.

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H-3 Backhoe boom - Midsection bail sheave shaft.
H-4 Other critical points of backhoe attachments.

*Recommended rosette gages.

(e) Procedure. The wire rope required for making the gantry test shall be supplied by the supplier. Strain test for each strain gage shall be repeated for at least three operating test cycles with applicable attachment. Superstructure tests shall be conducted in conjunction with crane-boom tests and with other applicable attachments as described herein only if the boom cable arrangement for these attachments is different than that required for the crane boom. In addition to the strain-gage tests, visual examination shall be made immediately after each test cycle to note any indication of yield or failure (this shall be construed to be any form of permanent deformation to include bending, weld failures, cracks, and scoring of bearings, pins, or bushings). Where rotation of drums, sheaves, or gearing becomes eccentric due to test loading, shafts shall be removed and checked with a straightedge for permanent deformation. Any zero shift in the instrument during a test cycle where failure of the instrument to return to zero after loading may be indication of yield along lines of principal stress other than the longitudinal axis of the gage.

(1) Pendant equalizer. Strain tests of the pendant equalizer shall be conducted while hoisting boom from ground, while hoisting, swinging, and lowering capacity loads (see 6.9.4) at minimum and maximum radii over the side or rear whichever is the most stable; and strain tests shall be conducted while booming out and in capacity loads at minimum rated radius to the tipping point and

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booming out and in a load which will induce tipping at the maximum rated radius. In addition, strain tests shall be conducted for those components not receiving their maximum stress in the most stable direction, while hoisting, swinging, and lowering capacity loads at minimum and maximum radii over the side or rear whichever is the least stable direction. Maximum strains shall be recorded during each testing cycle on the strain-gage locations described in 4.6.2.13(d). Swinging of capacity load over the side shall be through a cycle of 180 degrees and over the rear through a cycle of approximately 10 degrees on either side of the longitudinal axis. A 60-foot boom shall be lifted from ground level up to 45 degrees; strain-gage readings shall be taken on all superstructure and pendant equalizer components.

(2) Shovel-front attachment. Strain tests of the shovel-front attachment shall be conducted with the shovel boom at an angle of 45 degrees to the horizontal, the dipper stick at an angle of 45 degrees to the horizontal, and the bucket teeth hooked beneath a stationary object (simulating quarry excavation). Hoist the shovel to engine stall or incipient vehicle tip, whichever occurs first, and record maximum induced strains. This operation shall be conducted operating over the side and over the rear.

(3) Backhoe attachment. Strain tests of the backhoe attachment shall be conducted excavating in heavy undisturbed clay to maximum rated digging depth of the machine. Bucket retrieving shall be limited by engine stall or incipient vehicle tip, whichever occurs first.

Note: Strain gaging is not required on front-end attachments which are provided for testing only, either as Government-loaned property or furnished by the supplier at no cost to the Government. Strain gage testing is not required on crane booms that comply with MIL-B-52203.

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(f) Records. In order to provide uniform records, data shall be maximum stress recorded during static tests and maximum stress obtained from strip recorder during dynamic tests and shall be recorded as follows:

MAXIMUM WORKING STRESS

Contract Order No. _____ Place _____

Front-End Attachment _____ Date _____

Recording Instruments _____ Recorder _____

Gage Type _____

Gage Factor _____

Gage Resistance _____

Gage No.	Component	Gage	Location	Material 1/	Operation	Recorded Strain Microinches									Maximum 2/ psi 3/ (Max)
						Test 1			Test 2			Test 3			
						(a) Zero Reading	(b) Test Reading	(b-a) Strain	(a) Zero Reading	(b) Test Reading	(b-a) Strain	(a) Zero Reading	(b) Test Reading	(b-a) Strain	

1/ Modulus of elasticity (E).
2/ Where rosette gages are used, see manufacturer's instruction for exact strain.
3/ Stress = Actual Strain (XE).

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(g) Qualification. The yield strength and modulus of elasticity of the material of the structural member shall be obtained from "Engineering Alloys" published by the Reinhold Publishing Corporation, from other recognized standards, or from certified laboratory test of the material concerned. Maximum unit stress imposed by the required loading exceeding 60 percent of the yield point for any structural member shall constitute failure of this test.

4.6.2.14 Lifting capacity tests. Lifting capacities of crane-shovel shall be determined by use of actual weights or load cells using a standard length boom conforming to 3.9.1.1. With the boom and crane-shovel in the least stable direction determine the tipping load with the boom at the minimum working radius, the boom radius equal to boom length, and at three intermediate points (see 6.9.2 and 6.9.5). Overload indicating system may be disconnected during this test.

(a) Rated lifting load computation. A curve shall be plotted using the tipping loads and radii measured above as points on the curve. A curve representing 75 percent of the tipping loads obtained shall be plotted.
(b) Rated loads less than those shown in table II shall constitute failure of this test.

4.6.2.15 Craning. Utilizing various loads including rated loads determined in 4.6.2.14 perform craning operations for 50 hours. Inability to perform craning, inability to complete the test, malfunction, evidence of any damage, or failure of any crane component shall constitute failure of this test.

4.6.2.16 Crane-capacity plate determination. Repeat the procedure specified in 4.6.2.14 for each additional 10 feet of boom length up to 60 feet. Plot 75 percent of the load values determined for 5-foot increments of minimum rated radius up to boom length radius for each 10-foot increment of boom length. A capacity chart for the jib shall be included. The overload indicating system may be disconnected during this test.

4.6.2.16.1 Overload indicating system. Utilizing a 30 foot boom set at first for the minimum and then the maximum operating radius, perform the following tests:

1. Assemble a load of a size which will not activate the visual signal.
2. Lift the load 2 feet off the ground.
3. Increase the load by small increments and determine the load required to activate the visual signal.
4. Continue to increase the load and determine the load required to activate the audible signal.

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(5) Reduce the load and determine the load at which the audible signal ceases.
(6) Continue to reduce the load to determine the load at which the visual signal stops.
(7) Lower the load to the ground and repeat the above procedure for the 40, 50, and 60 foot booms.
(8) Failure of the overload indicating system and its components to meet the requirements of 3.15.1.4.1, inability to complete the test, or malfunction shall constitute failure of this test.

4.6.2.17 Rope strength. Measure the maximum load on all ropes at 30-feet operating radius and rated loads and while raising a boom 60 feet long from the ground. A factor of safety less than specified in 3.15.1.6 shall constitute failure of this test.

4.6.2.18 Power and speed test. This test shall be conducted at an actual or simulated altitude of 5,000 feet. If the option is taken to simulate altitude conditions for test purposes, such simulation shall be accomplished in the following manner. In the vicinity of the engine, mount a container of sufficient capacity and inlet orifice size to assure a steady air pressure to the engine equal to 24.89 inches of mercury. Connect the outlet of the container to the air cleaner so that air is inducted into the engine in the normal manner through the air cleaner filter. Determine the intake air pressure with a U-tube manometer tapped into the container at a point where static air pressure is maintained in the container. The intake and exhaust systems shall be complete and unaltered. Exhaust back pressure shall not be reduced for this test.

(a) Assemble a test load of not less than 15,625 pounds and boom at minimum radius.
(b) By trial lifts, adjust the load so as to require two parts of hoist line and provide steady hoisting rate near governed engine speed. Single sheave block may be used. Establish a mark on the hoisting rope at a point near the primary hoist drum where the load is 3 feet clear of the ground. Establish a second mark on the rope when the load has been lifted to a point where the blocks are approximately 3 feet apart or before interference occurs between the load and crane structure. Lower the load and measure the lineal feet of rope between the two marks. Test shall be started with not less than three wraps on the hoist line drum.
(c) Using a stop watch accurate to plus or minus 0.2 second in 1 minute, time the interval required for the two marks on the hoisting rope to pass a selected point while the load is being lifted with the engine operating at governed speed. At least three checks shall be made and averaged.
(d) Compute the hoist-line speed in feet per minute. Multiply this computed line speed by the load weight in pounds divided by the number of parts of hoisting line.

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(e) Assemble a test load equal to the line pull determined by conditions specified in 3.12.1 (line power in foot-pound per minute divided by the corresponding minimum line speed in feet per minute). Lift this load with a single hoist line, and simultaneously rotate the superstructure.

(f) With basic length of boom, rest boom tip on ground to insure that three wraps of cable are left on boom hoist drum (see 3.15.1.6.2).

(g) Evidence of any of the following shall constitute failure of this test:

(1) Hoisting drum line speed and hoisting drum line power not within limits specified in 3.12.1.
(2) Failure to lift the specified load and simultaneously rotate the superstructure.
(3) Less than three wraps of cable on boom hoist drum with boom tip on ground.

4.6.2.19 Swing. Preliminary to this test, check full-load engine speed to see that it does not exceed governed speed as specified in 3.13.1.

(a) With a load equal to 50 percent of maximum line pull at 160 feet per minute and the boom at a 40 degree angle, time with a stop watch 10 revolutions of the loaded superstructure at full throttle. Repeat twice. Swing speed less than 3.5 rpm shall constitute failure of this test.

(b) With the crane superstructure loaded as specified in (a) above, rotate the superstructure at full throttle for 10 minutes in each direction to determine heat in rollers. Thermocouples or contact pyrometer may be used to determine temperature. A temperature of more than 175° F. shall constitute failure of this test.

(c) With the attachments, loading, and radius as specified in 3.12.8.3, shall make not less than three and one-half 90 degree passes per minute with the superstructure for 1-1/2 hours. Thermocouples or contact pyrometer may be used to measure the temperature in the swing clutches. A maximum temperature of more than 500° F. or fading shall constitute failure of this test.

4.6.2.20 Power lowering. Inspection shall be made of the power-down system for lubrication, ease of operation of controls, safety precautions, and for workmanship.

(a) The crane shall be equipped with a boom of appropriate length for the power-load-lowering tests.

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(b) Position the crane superstructure in the most stable craning position, and the boom reeved with a load line. The test load assembled and used for this test shall weigh, not including the block and cable, the same ratio of the maximum capacity load (see 6.9.4) as the ratio of the parts of load line to the number of lines used in tipping tests (see 4.6.2.14) (e.g., if a 4-part load line is standard reeving with a capacity load, a 2-part load line will be used with a 1/2 capacity load).

(c) Position the boom for a load radius of 12-feet plus or minus 2-inches.

(d) With the hoist drum brake unlocked hoist the load 20 feet plus or minus 3 inches above the ground.

(e) Adjust the hoist drum brake to hold a maximum capacity load, and apply the hoist drum brake until the brake will hold the load suspended.

(f) Prepare the crane for power-down operation on the load line.

(g) With brake set and locked for holding the load and the engine set at an idle speed, bring the power-down clutch into full engagement. If this stalls the engine, increase the idle speed until the engine will "ride-over" the hoist brake.

(h) As soon as the load reaches the ground, immediately repeat operations specified in (d), (f), and (g) above.

This cycle shall be repeated 15 times if a 4-part line and the maximum capacity load is used or 30 times if a 2-part line and one-half the maximum capacity load is used.

(i) Check the temperatures of all brake linings employed in this operation with a contact pyrometer or at least four thermocouples imbedded within each lining near the working face and uniformly spaced around the perimeter.

(j) Assemble a load equal to one-half maximum-capacity load, and repeat tests specified in (c) through (i) above.

(k) By the use of a tachometer on the engine and a tachometer on the main hoist drum shaft, check the ratio of the engine speed to the main hoist drum shaft.

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Failure to override the main hoist drum brake; temperatures of the brake linings exceeding 600° F. or the maximum recommended by the manufacturer of the lining for the application, whichever is less; main hoist drum speed greater than 50 percent or less than 30 percent of the engine speed; inability to accomplish power lowering without pumping the control lever; evidence of leakage, corrosion, any malfunction, permanent deformation, or breakage shall constitute failure of this test. After completion of this test, remove the brake bands and examine the brake linings for evidence of imminent or impending failure due to (1) cracking, (2) internal rupture (indicated by distortion of rivet holes), or (3) wear (any amount beyond crazing of surface not to exceed 1/16-inch). Evidence of damage or weakness described above shall constitute failure of this test.

4.6.2.21 Clutch capacity. With a 30-foot boom and at a short radius, hoist to a height of 5 feet above the ground a load that will develop a line pull in pounds equal to the maximum line pull developed by the hoist drum. Apply the hoist-drum brake to hold the load. Stop the engine and jam lock a meshing set of gears between the engine and the power-down clutch. Engage the power-down clutch for load lowering, release the hoist-drum brake, and let the load be supported on the power-down clutch which in turn shall be fixed by virtue of the locked gears. As the brake is released, examine the power-down clutch for slippage. Any slipping of the clutch, malfunction, permanent deformation, or breakage shall constitute failure of this test.

4.6.2.21.1 Backing down. Suspend on a single line a load that will develop the single-line pull equal to 133 percent of rated load at recommended reeving; stop the engine; jam lock a meshing pair of gears; knuckle-in the primary hoist drum clutch; and release the brake holding the load. After the brake is released, inspect the clutch for slippage. Any clutch slippage shall constitute failure of this test.

(a) Repeat the backing down procedure using the secondary hoist drum clutch.

(b) Suspend the same load as used above on a single line; stop the engine; knuckle-in both the secondary and primary drum clutches; disengage the master clutch; and lower the load on the brake backing down the hoisting machinery. An inspection shall be made for ease of control operation and smoothness of load lowering and machinery operation. Any binding, chattering of machinery, slipping of the clutches, or unusual effort required for the operation shall constitute failure of this test.

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4.6.2.21.2 Clutch control. When a power hydraulic control system is furnished for the superstructure (see 3.12.13.3) the following test shall be performed. Without engine power, a clutch shall be engaged and released for not less than 15 cycles. Inability of the clutch to engage and release at least 15 times using only the accumulator pressure, without engine power, shall constitute failure of this test.

4.6.2.22 Boom hoist cable angle. With the boom at an angle of 82 degrees from the horizontal, measure the fleet angle made by boom hoist cable with all sheaves on the A-frame and pendant equalizer where a divergence is made by the cable across the cable groove. The measurement shall be made with a metal template not less than 5 feet in length. Any fleet angle greater than 6 degrees shall constitute failure of this test.

4.6.2.23 Boom stop and pile lead.

(a) Calculate the load imposed on the boom stop and attaching points when bending a boom over the stops. Subject the boom stop to this load. The basic boom shall be lowered until the boom point touches the ground; then raise the point until the boom makes an angle of not less than 25 degrees with the horizontal. Repeat this cycle for a total of not less than 12 times, all at maximum boom hoist speed.

(b) The pile-driving leads, boom adapter and catwalk specified in 3.15.2.8 or 3.29 shall be completely attached to the boom and revolving superstructure and be ready for pile-driving operation. The mounting shall be checked.

(c) Failure of boom stop to withstand force required for completely bending the boom, failure of boom stop to stop boom at 82 degrees, any noticable binding or chattering of the boom stop when raising boom or incorrect fitting or mounting of the pile-driving leads shall constitute failure of this test.

4.6.2.24 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.12.12.2. The supplier shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

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4.6.2.25 Counterweight removal. Remove the counterweight from the crane-shovel and place it on a truck or trailer. Reverse the operation and replace the counterweight to its original position. This test shall be accomplished by not more than two men using only those lifting provisions of the crane-shovel. Use of equipment or machinery other than as specified above or inability to perform the removal or replacement of the counterweight in a time limit of 30 minutes shall constitute failure of this test.

4.6.2.26 Backhoe and shovel-front.

4.6.2.26.1 Backhoe. Perform maximum capacity heavy-duty trenching operations in undisturbed heavy clay soil at minimum depths specified in table IV for not less than 40 hours. The crane-shovels shall average not less than 60 cu. yd. per hour. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.6.2.26.2 Shovel-front attachment. Perform shovel-front excavating operations at maximum capacity and include equal portions of time operating at each of the ranges specified in table III. The total period of operation shall be not less than 40 hours. The operations shall be conducted in a gravel pit or rock quarry. The crane-shovel shall average not less than 60 cu. yd. per hour. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.6.2.27 Clamshell and dragline. Perform maximum-capacity, heavy-duty clamshell and dragline operations in undisturbed heavy clay soil. The operation shall be performed with the clamshell and dragline bucket for at least 16 hours (8 hours for each bucket). The crane-shovel shall average not less than 55 cu. yd. per hour. Inability to operate with clamshell and dragline buckets, inability to complete test, malfunction, evidence of any damage or failure of any component shall constitute failure of this test.

4.6.2.28 Climatic.

4.6.2.28.1 Operating temperature.

4.6.2.28.1.1 High-temperature operation. Subject the nonwinterized crane-shovel to an ambient temperature of 125° F. until the temperature

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of the crane-shovel has stabilized. Start the engine and operate it at maximum governed speed for a total of 1 hour. During this hour, individually engage and disengage clutches instantly under maximum governed engine speed. Actuate the brake controls. Move the superstructure clockwise and counterclockwise and the crane forward and backwards within the limitations of available space. Stop the engine at the end of 1 hour; restart the engine after 5 minutes and operate for 10 minutes without load. Failure of the engine to start and operate within the limits of applicable engine specifications or failure of any component to function shall constitute failure of this test.

4.6.2.28.1.1.1 Thermocouple location. The crane-shovel shall have not less than the following thermocouples installed prior to the high temperature test.

(a) Engine coolant temperature: One in the engine-cooling-jacket discharge pipe and another in the coolant inlet to the engine.
(b) Fuel temperature: In the tank.
(c) Battery temperature: In the two center cells of each battery between the plates and at mid-depth of the electrolyte.
(d) Ambient temperature measured at an angle of 45 degrees and not less than 3 feet from each corner of the crane-shovel at engine-crankshaft level.
(e) Engine oil gallery.
(f) Torque converter sump. (When torque converter is furnished).

Record at 10-minute intervals temperature readings of the thermocouples listed above during the operations specified in 4.6.2.28.1.1.

4.6.2.28.1.2 Low temperature. The following test procedure shall apply and all data shall be recorded for the low-temperature operation (minus 25° F.). The crane-shovel shall have installed, prior to the test, thermocouples as listed in 4.6.2.28.1.1(a) through (f) and temperatures shall be recorded prior to engine starting, immediately after engine starting, and at 10-minute intervals during engine warm-up and cycling.

(a) Check the specific gravity of each battery to insure full charge.

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(b) Subject the crane-shovel to an ambient temperature of minus 25° F. or lower until all parts, lube oil, coolant, fuel and batteries have stabilized at minus 25° F. or lower.

(c) Maintain an ambient temperature of minus 25° F. or lower throughout the entire test.

(d) Start the crane-shovel engine. Engine should run on diesel fuel in not more than 5 minutes after beginning of first cranking.

(e) One operator wearing arctic clothing including mittens shall perform all priming and starting operations.

(f) Crank the engine using batteries that were cold soaked and stabilized at minus 25° F. or lower.

(g) Substitute or supplementary batteries or other electrical cranking energy shall not be used.

(h) Glow plugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the crane-shovel and normally used in starting the engine may be used.

(i) Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained within 15 minutes after starting.

(j) When started, run engine 15 minutes. After completion of the 15-minute operation, operate the engine as follows:
   (1) 10 minute idling.
   (2) 10 minutes half throttle.
   (3) 5 minutes full throttle.

   Perform the 25-minute cycle three times.

(k) During engine cycling and while wearing arctic mittens and boots, the operator shall perform each of the following:
   (1) Engage and disengage power train to the crawler, steering and brake mechanism.
   (2) Operate through two complete movement cycles all brakes, shafts, and cylinders.
   (3) Mount to and dismount from the operator's position on the crane-shovel.

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(1) Determine functioning of the crane-shovel. Individually engage and disengage superstructure clutches under maximum governed engine speed. Actuate brake controls. Move superstructure clockwise and counterclockwise within limitations of available space. Drive the crane-shovel forward and backward within limitations of available space. Failure of the engine to start and operate within the limits of applicable engine specification or failure of any component to function shall constitute failure of this test.

Upon completion of the cycling period, shut down the engine and (1) while the crane-shovel is still in the chamber at minus 25° F. or lower, and (2) while the operator is wearing arctic mittens, restart the engine by using outside power through the slave receptacle. Failure to start the engine with outside power through the slave receptacle shall constitute failure of this test.

4.6.2.28.2 Storage.

4.6.2.28.2.1 High-temperature storage. Subject the crane-shovel to a high ambient temperature until all parts have stabilized at plus 160° F. The temperature of all parts shall then be maintained continuously at plus 160° F. for 1 hour. Examine the crane-shovel visually and without disassembly. Any evidence of damage, leaks, cracks, permanent deformation, or discoloration shall constitute failure of this test.

4.6.2.28.2.2 Low-temperature storage. Subject the crane-shovel to a low ambient temperature of minus 50° F. until all parts have stabilized at minus 50° F. The temperature of all parts shall then be maintained continuously at minus 50° F. for at least 6 hours. Examine the crane-shovel visually and without disassembly. Any evidence of damage, leaks, cracks, permanent deformation, or discoloration shall constitute failure of this test.

4.6.2.28.3 Rain. Subject the crane-shovel to a simulated rainfall of 4-inches plus or minus 1-inch per hour as measured by a U.S. Weather Bureau type gage. The simulated rainfall shall fall on the crane-shovel at an angle of 45 degrees, plus or minus 5 degrees and impinging with a force equivalent to a 20 mile an hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the crane-shovel simultaneously for not less than 30 minutes on each surface. The engine shall

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be started (without removing accumulated water) 26 minutes after beginning the rain test, and idle for the remainder of the test. At the completion of the test (within 30 minutes) the door shall be opened, the inside of the cab shall be examined, and all craning functions shall be performed (including lifting and hoisting a capacity load). Evidence of water penetration such as free water in the operator's area and machinery area, inability of the engine to start and idle, or inability to perform craning functions as specified herein shall constitute failure of this test.

4.6.2.28.4 Winterization. In addition to the tests specified in table XI (excluding the 125° F. test) the winterized crane-shovel shall be subjected to the following tests as indicated by an "X" in table XII.

4.6.2.28.4.1 High-temperature operation. Subject the crane-shovel to an ambient temperature of 105° F. until the temperature of the crane-shovel has stabilized. The crane-shovel shall be tested with the winterfronts in open position and the engine side panels removed. Start the crane-shovel engine and operate it at maximum governed speed for a total of 1 hour. During this hour, individually engage and disengage clutches instantly under maximum governed engine speed. Actuate the brake controls. Move the superstructure clockwise and counterclockwise within the limitations of available space. Stop the engine at the end of 1 hour; restart the engine after 5 minutes and operate for 10 minutes without load. Failure of the engine to start and operate within the limits of applicable engine specifications or failure of any component to function shall constitute failure of this test.

4.6.2.28.4.2 Cold starting and running. The crane when winterized shall be tested in accordance with test method No. 1. One or more of the following shall constitute failure of this test:

(a) Engine does not start within 5 minutes after 4 hours of standby heat.
(b) Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without the use of primers and priming fluids.
(c) Engine oil sump temperature lower than plus 50° F. prior to engine starting.
(d) Battery temperature lower than plus 50° F. prior to engine starting.

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(e) Engine coolant temperature at heater inlet lower than plus 50° F. prior to engine starting.
(f) Battery compartment temperature less than plus 50° F. subsequent to engine start.
(g) Cab temperature less than plus 35° F. at operator's foot pedals within 1 hour after starting engine.
(h) 75 percent of each window and windshield not frost-free within 1 hour after starting engine.
(i) Lubricating oil temperature not stabilized in the engine lubricating oil gallery within the range of 130° to 230° F. within 1 hour after starting and remain within this range throughout the 1-hour test period.
(j) Coolant temperature not stabilized in the engine-cooling jacket discharge pipe within the range of 140° to 210° F. within 1 hour after starting and remain within this range throughout the 1-hour test period.
(k) Cab temperature at operator's foot pedal after cab heaters have operated 1 hour could not be maintained at plus 35° F. or warmer.
(l) Instruments and gages do not function properly.
(m) Thermal damage to batteries.
(n) Battery charging system does not show positive charge.
(o) While the operator is wearing arctic mittens and boots:
    (1) Power train cannot be engaged to the wheels.
    (2) All clutches, brakes, shafts, drums, and hydraulic cylinders cannot be operated through two movement cycles.
    (3) Windshield wipers cannot be manually and power operated.
    (4) All cab doors, engine housing doors, and winterfronts cannot be operated, latched, and unlocked.
    (5) Operator cannot safely mount to the operator's position in the superstructure and cannot safely dismount from the operator's position.
    (6) Maintenance checks cannot be performed.

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4.6.2.29 Post test disassembly and examination. Upon completion of all tests, the crane-shovel shall be dismantled and the following components shall be disassembled in detail for a visual examination:

(a) Engine. The engine shall be removed from its mounting and disassembled in detail as follows: Remove the cylinder heads, all valves, the two pistons and piston pins from the two cylinders having the lowest pressure reading at the end of the tests. Remove two main bearings and two connecting rod bearings and turbocharger if one is provided with the engine.
(b) Torque convertor (if furnished).
(c) Brakes and clutches.
(d) Ring gear and pinion assembly.
(e) Track and turntable rollers.
(f) Hydraulic system.
    (1) Accumulators
    (2) Cylinders
    (3) Hose and fittings where leakage has occurred.
(g) Drums including appropriate clutch and brake operating parts.
(h) Winterization system (when furnished).

Additional disassembly and examination shall be made at the option of the contracting officer. A complete description of the condition of all components and parts of the crane shall be recorded and photographs shall be taken of any deficiency. Any evidence of fretting, overheating, permanent deformation, or any wear of parts, or parts of components, or wear exceeding manufacturers published tolerances shall constitute failure of this test.

4.6.2.30 Individual tests. Each crane-shovel shall be equipped with a 30-foot boom except as specified herein, and shall be tested as follows. Inability to perform any of the following operations or evidence of any malfunction shall constitute failure of any test specified herein.

4.6.2.30.1 Brake and clutch test. Perform all hoisting and swinging functions for a period of 30 minutes with a 6000 pound load and with the hoist and swing brakes partially set.

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4.6.2.30.2 Swing test. Without test load revolve the superstructure clockwise for 15 minutes; repeat the test counterclockwise.

4.6.2.30.3 Hoisting test. With a 6,000 pound load at a 20-foot radius on a single line attached to the primary hoist drum: (a) Hoist and lower on the brake for 50 cycles; (b) hoist and power lower for 50 cycles. Repeat these tests with the line attached to the secondary drum.

4.6.2.30.4 Boom hoisting. Raise a 60-foot boom from ground to an angle of 60 degrees from the horizontal and lower to ground level for 10 cycles. The supplier, at his option, may perform this test with a 30-foot boom and a hook load with a combined total weight not less than the 60-foot boom.

4.6.2.30.5 Boom-stop test. Hoist the boom to engage the boom stop. Failure of the boom stop to stop the boom at 82 degrees shall constitute failure of this test.

4.6.2.30.6 Off road. Each completely assembled crane-shovel shall be off-road tested for a distance of not less than 1 mile. This 1 mile shall include the following:

(a) One grade not less than 30 percent from the horizontal, not less than 25 feet long, and the crane-shovel shall be driven uphill and downhill over this grade.
(b) The crane-shovel shall be driven backward not less than 200 feet.
(c) The brake shall be tested by parking on the 30 percent grade.

Failure of the crane-shovel to accomplish any part of this test shall constitute failure of this test.

4.6.2.30.7 Overload system. Perform the test specified in 4.6.2.16.1 using the minimum operating radius and a 30-foot boom. Failure of the overload system to operate as specified in 3.15.1.4.1 shall constitute failure of this test.

4.6.2.31 Sound level measurement.

4.6.2.31.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise stationary equipment at the operating condition of governed engine speed lifting two-thirds of rated load on a level test surface.

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4.6.2.31.2 Representative duty position. Measure the sound level at eight evenly spaced points, 68 inches above ground level, on a circle of 25 foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The operating condition shall be governed engine speed lifting two-thirds of rated load on a level test surface.

4.6.2.31.3 Exterior spectator. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise.

4.6.2.32 Cooling test. The crane-shovel shall be tested in accordance with Test Method No. 2. Extrapolated water temperature exceeding 210° F. leaving engine cooling jacket, inability to maintain extrapolated engine temperature within the range specified in MIL-E-11276, or failure of the controls and instruments to operate shall constitute failure of this test.

4.6.2.33 Mean-time-between-failure. Using the MTBF specified in 3.8.5, the initial production crane shall be tested in accordance with Test Plan II of MIL-STD-781. Test hours or miles in 4.6.2.10.5, 4.6.2.10.6(b), 4.6.2.15, 4.6.2.26.1, 4.6.2.26.2, and 4.6.2.27 shall be reduced to 94 percent of the values specified for the first assessment of MTBF. The road milage shall be divided (computed based on speed of 0.75 mph) and a part conducted between each of the operations. The "specified" mean-time-between-failure shall be accepted if the test times and the failures fall within the "accept" region and rejected if it falls within the "reject" region of Test Plan II, and testing shall be terminated. If the assessment falls within the "continue test" region, multiples of the previous testing shall be performed until an "accept" or "reject" assessment is achieved. Failure for the purpose of this test is defined as any random malfunction (dependent malfunctions are excluded) which causes degradation below the operating condition required to continue tests specified above and cannot be remedied within 1 hour (all waiting periods are excluded) utilizing the resources available to the using unit. Nonconformance to 3.8.5 shall constitute failure of this test.

4.6.2.34 Maintenance evaluation.

4.6.2.34.1 Maintenance ratio. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.6. The ratio of all the manhours of maintenance required to the hours of operation performed shall be computed. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.2.34.2 Maintenance package. During initial production testing, maintenance support elements such as manuals (when furnished), lubrication charts, repair parts and special tools (see 3.17), shall be evaluated. Failure of the crane to be properly maintained utilizing the support elements shall constitute failure of this test and shall require revision of the document(s). Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the mean-time-between-failure (see 4.6.2.34).

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4.6.2.35 Torque-converter stall. The torque-converter shall be stalled at full throttle of the crane engine for not less than 30 seconds. The crane shall then be moved for a period of 15 minutes. Oil temperature exceeding 250° F. from an initial temperature of not less than plus 180° F. in an ambient temperature of not less than 30° F. or inability of the torque-converter oil to return to a stabilized operating temperature within 15 minutes shall constitute failure of this test. This test may be conducted during the cooling test specified in 4.6.2.32.

4.7 Inspection comparison. The Government may select crane-shovels at any time during the contract production period and subject these crane-shovels to the examination identified by X in column 5 of table IX and the test identified by X in column 3 of table XI to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production crane-shovels. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison crane-shovel to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison crane-shovel(s) and shall be cause for refusal by the Government to continue acceptance of production crane-shovels until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on crane-shovels previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all crane-shovels accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with MIL-C-3580.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The crane-shovels, attachments, repair parts, tools, and technical publications shall be preserved, packaged, packed, and marked in accordance with MIL-C-3580. The level of preservation, packaging and packing shall be level A, B, or C, as specified.

6. NOTES

6.1 Intended use. The crane-shovels, together with the various items of front-end operating equipment, are intended for use in handling materials, lifting and transporting loads short distances, soil stripping, earthmoving ditch digging, and pile driving.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of preproduction model cranes, time frame required for submission and where test(s) shall be performed (see 3.2.1).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2.1).
(d) When initial production testing is required, number of initial production cranes to be furnished, and where test(s) shall be performed (see 3.3).
(e) When a hydraulic torque converter is required (see 3.12.11).
(f) Composition required (see 3.12.12.1).
(g) When primary front end attachments are not required as specified (see 3.15.1).
(h) When an overload indicating system is required (see 3.15.1.4.1).
(i) When boom extensions are required (see 3.15.2.1).
(j) When boom jib is required (see 3.15.2.2).
(k) When clamshell bucket is required (see 3.15.2.3).
(l) When clamshell bucket is to be furnished with side and corner teeth (see 3.15.2.3).

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(m) When tag-line is required (see 3.15.2.4).
(n) When dragline bucket is required (see 3.15.2.5).
(o) When dragline fair-leads are required (see 3.15.2.6).
(p) When wrecking balls are required and size of wrecking balls to be furnished (see 3.15.2.7).
(q) When pile-driver leads, boom adapters and catwalks are required (see 3.15.2.8).
(r) When piledriver hammers are required, drop or diesel, and diesel size required (see 3.15.2.9).
(s) When shovel front is required (see 3.15.2.10).
(t) When backhoe is required (see 3.15.2.11).
(u) When winterized crane-shovels are required (see 3.16).
(v) Grades of oils required at time of delivery if other than that specified in 3.18.
(w) When vehicle class sign kit is not required (see 3.24.2).
(x) When a rifle mounting kit is required (see 3.27).
(y) What Government-loaned property will be furnished (see 3.29).
(z) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production crane-shovels from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish crane-shovels conforming to this specification.

6.4 Welding of high strength constructional steels. When high strength constructional steels are used in the fabrication of the crane-shovel, the contracting officer should require the supplier to include in the technical publications specified (see 3.2.3 and 6.11) the identification and location of these steels, and the necessary precautions to be taken by the using agency in the repair and replacement of these steels. These precautions will include but not be limited to:

(a) How to avoid excessive restraint in a weld joint that could produce weld-metal or heat-affected-zone cracking.

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(b) Type of weld filler metal to be used for repair of each different type of steel by the shielded Electric-Arc (coated electrode) process and by the Metal Inert Gas (MIG) process including:
  (1) Type and class of electrode and MIG wire.
  (2) Size (diameter) of electrode and MIG wire.
  (3) Shielding medium for MIG process.

(c) Material handling safeguards for coated electrodes and MIG wire to prevent contamination and moisture absorption, including the treatment of coated electrodes and MIG wire after exposure to contamination and moisture absorption.

(d) Preparation of base metal for welding. Method to be used (grinding or other). Preparation of root for welding second side when thickness of material necessitates that joint be welded from both sides.

(e) Base metal temperatures.
  (1) Minimum temperature of base metal at start of welding.
  (2) Minimum and maximum preheat of base metal at start of welding.
  (3) Maximum interpass temperature during welding.

(f) Rate of flow of shielding medium for MIG process.

(g) Maximum heat input during welding.

(h) The minimum ambient temperature at which welding should be done.

(i) Post weld heat-treatment (stress-relieving) requirements (necessary when weldment must retain dimensional stability during machining or stress-corrosion is a factor).

6.5 Information figures. Figures 4 and 5 show a type of labyrinth seal which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other labyrinth seal shapes which conform to this specification.

6.6 Lubricants and oils. The contracting officer should furnish a list of military lubricants applicable to the crane-shovel covered by this specification as contained in the Federal Supply Catalogue, Department of Defense, Section, Identification List C9100-IL, for FSC Group 91.

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6.7 Government-furnished property. The contracting officer should arrange to furnish the supplier the property listed in 3.27.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.29 unless the item is being purchased under the same contract.

6.9 Definitions.

6.9.1 Boom radius. The boom radius is the horizontal distance between the centerline of rotation of the crane-shovel superstructure and the centerline of the hook block.

6.9.2 Tipping. The tipping point is that condition where the crawler rollers on the side opposite the load are lifted from contact with the treads, when lifting over the side, or where 1/3 of the bearing area of properly adjusted treads are lifted from the ground, when lifting over front or rear.

6.9.3 Rated load. Rated load shall be a load that is not more than 75 percent of the tipping load for a given radius with the crane resting on firm, level ground, fully serviced, and with the fuel tanks not less than one-half full.

6.9.4 Capacity loads. Capacity loads are loads which are not less than 133 percent of the loads shown in table II.

6.9.5 Tipping load. Tipping load is the weight required to cause tipping at any specified radius. It shall consist of the hook block slings and other load weight components attached to the hook of the hook block and shall not include weight of the crane boom wire rope, hook block or other components necessary for crane operations.

6.9.6 Gross weight. Gross weight includes the complete crane-shovel, basic crane boom, crane hook block, complete accessories, lubricants and coolant necessary to completely service the crane-shovel for operation, and fuel tanks at least half-full.

6.10 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the supplier and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the crane. The supplier should then be requested by the contracting officer to apply the weight classification number(s) to the crane in accordance with AR 746-5. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.24.2).

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6.11 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists and supplier's maintenance and operation manual to be furnished with each crane.

6.12 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools, and instructions on shipment of cranes. A suggested paragraph is as follows:

> "Shipment of cranes shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.13 The contracting officer should arrange to furnish the Warning Decals, Part Number 385-101(19099) and provide mounting instructions as required in 3.22.1.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 3810-A039

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APPENDIX

FOR

CRANE-SHOVEL, CRAWLER MOUNTED, 12-1/2-TON AT 12-FOOT RADIUS, 3/4 CUBIC YARD, DIESEL-ENGINE-DRIVEN, FOR MILITARY OPERATIONS

Scope. The test methods contained in this appendix is to determine whether crane-shovels procured under this specification conform to the requirements set forth.

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TEST METHOD NO. 1

WINTERIZATION TEST (MINUS 50° F.)

1. TEST APPARATUS:

(a) Cold chamber.
(b) Arctic clothing.
(c) Thermocouples.
(d) Battery hydrometer.
(e) Wattmeter.
(f) Voltmeter.
(g) Recording instruments (temperature readings), electrical.

2. TEST PROCEDURE:

(a) Service the crane with arctic antifreeze conforming to MIL-C-11755 and with arctic lubricants.
(b) Check the specific gravity of each battery. Specific gravity shall be not lower than 1.275 before beginning of cool down.
(c) During the 4 hour standby heating and starting period, close and latch the housing panels. All operations incident to closing and covering up the engine shall be performed by one operator wearing arctic mittens.
(d) All operations incident to starting and priming the engine shall be performed by one man wearing arctic mittens.
(e) Subject the crane to an ambient temperature of minus 50° F. until all parts of the crane have stabilized at minus 50° F. Perform the test after stabilization is assured by temperature readings of minus 50° F. from thermocouples located as follows:

(1) In the two center cells of each battery between the plates and at mid-depth of the electrolyte.
(2) In the volumetric center of any hydraulic oil tank of more than 25-gallon capacity.
(3) In the engine lubricating oil gallery (at downstream end).
(4) In the engine coolant in the engine-cooling-jacket discharge pipe.

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(5) In the chamber at a horizontal angle of 45 degrees, from the sides and ends of the crane and not less than 3 feet from each corner of the crane at engine-crankshaft level.
(6) In the operator's compartment 6 inches above the floor boards and between the operator's foot pedals.

(f) Maintain minus 50° F. ambient temperature throughout the test.
(g) Perform preheating by interconnecting the crane to outside 115 volt power source using the 25 foot cable supplied with the crane for 4 hours. Start coolant-type cab heaters not more than 15 minutes after the engine has been started.
(h) Substitute or supplementary batteries or other electrical cranking energy shall not be used.
(i) Glow plugs or other accessories working off of the cranking batteries may be used in the minus 50° F. start. Use only the cold starting aids specified in 3.16.10.
(j) Start crane engine within 5 minutes after initial 4 hours of standby heat.
(k) Operate the engine until:

(1) The lubricating-oil temperature is stabilized in the lubricating oil gallery within the range of 130° to 230° F., and
(2) The coolant temperature is stabilized in the engine cooling-jacket discharge pipe within the range of 140° to 210° F.

(l) Run the engine for 1 hour after these conditions have been met.
(m) Cycle the engine periodically between idling and full load governed speed during the 1-hour period of engine operation specified in (1) above.
(n) Record at 10-minute intervals temperatures during 4 hour standby heating period, immediately after engine starting, during engine warmup, and during temperature stabilization and 1-hour operating period, including but not limited to the following:

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(1) Batteries.
(2) Hydraulic tank.
(3) Engine oil.
(4) Engine coolant.
(5) Ambient temperature.
(6) Operator's compartment 6 inches above the floor at the foot pedal.
(7) Converter oil at outlet (if furnished).

(o) During 1-hour operation and while wearing arctic clothing, an operator shall perform each of the following:

(1) Operate through two complete movement cycles all clutches, brakes, shafts, drums, cylinders, and hydraulic components.
(2) Operate windshield wipers.
(3) Operate all cab doors, engine housing doors, sidehoods, and winterfronts by opening, closing, latching, and unlocking each of these items.
(4) Mount to and dismount from the operator's position on the superstructure.

(p) Upon completion of the 1-hour period, shut down the engine, and while the crane is still in the chamber at minus 50° F., and the operator is wearing arctic mittens, he shall perform each of the following maintenance checks and operations:

(1) Restart the engine by using outside power through the battery-charging receptacle. Disconnect cable to outside power and then shut down the engine.
(2) Check oil level on all dipsticks.

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TEST METHOD NO. 2

COOLING TEST

1. TEST CONDITIONS:

The crane-shovel, with or without boom, shall be tested in an area where the change in the ambient air temperature shall not exceed plus 5° F. during a 4-hour period. A dynamometer or other loading device shall be directly connected to the drum shaft or chain connected to a sprocket on the drum. The test shall be conducted with doors and windows in the open position and access doors to the engine compartment latched open. The superstructure shall be serviced with military oils and greases designated for use in the ambient temperatures at which the test is to be run. The coolant system shall be serviced with water.

2. PROCEDURE:

(a) Instrumentation. Install thermocouples to measure the following temperatures:

(1) Air, ambient.
(2) Air, entering inlet grille.
(3) Air, leaving outlet grille.
(4) Air, within engine compartment.
(5) Air, entering air cleaner.
(6) Oil temperature, engine sump.
(7) Oil temperature, engine gallery.
(8) Water entering radiator (top tank).
(9) Water leaving radiator (bottom tank).
(10) Oil torque converter sump (if applicable).

(b) The ambient air temperature shall be measured at the engine crankshaft level, 4 feet from the external face of the radiator on suction type fan blade installation and 4 feet from the largest accessible opening to engine compartment on blower type fan blade installation.

(c) Start the engine and run for 10 minutes at low idle engine speed and record for 10 minutes the temperature readings of the thermocouples listed in (2).

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(d) With the engine at wide open throttle, with the recorder reading temperatures at not more than 3 minute intervals, load the loading device to obtain an equivalent to capacity load as specified in 6.9.4. Maintain this condition and record temperatures until oil and coolant temperatures have stabilized. Stabilization shall be considered achieved, when the temperature differential between the top tank and ambient air does not vary more than 2° F. for any 30 minute period.

(e) Extrapolate all temperature readings on a degree for degree straight line based on a 125° F. ambient.

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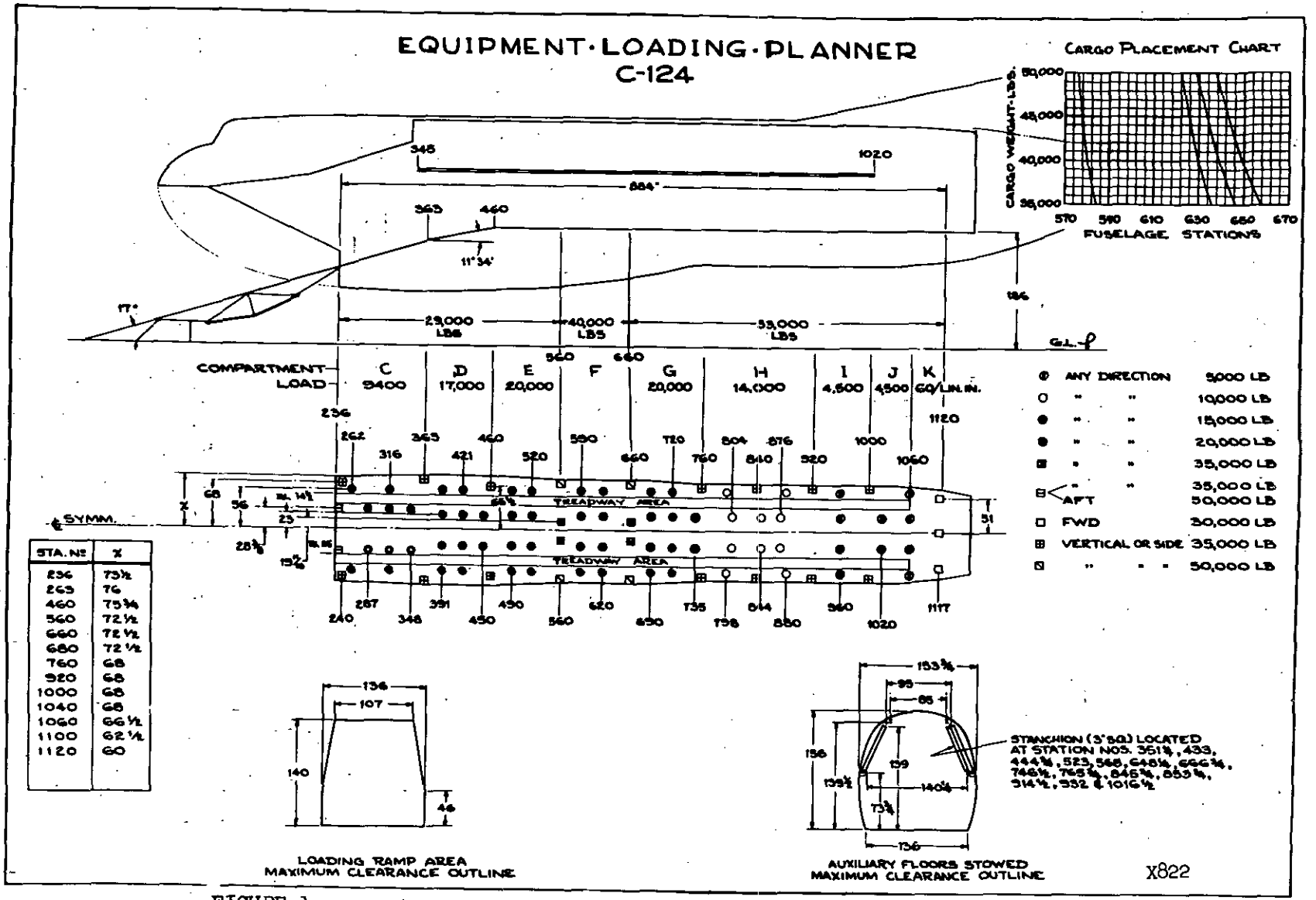
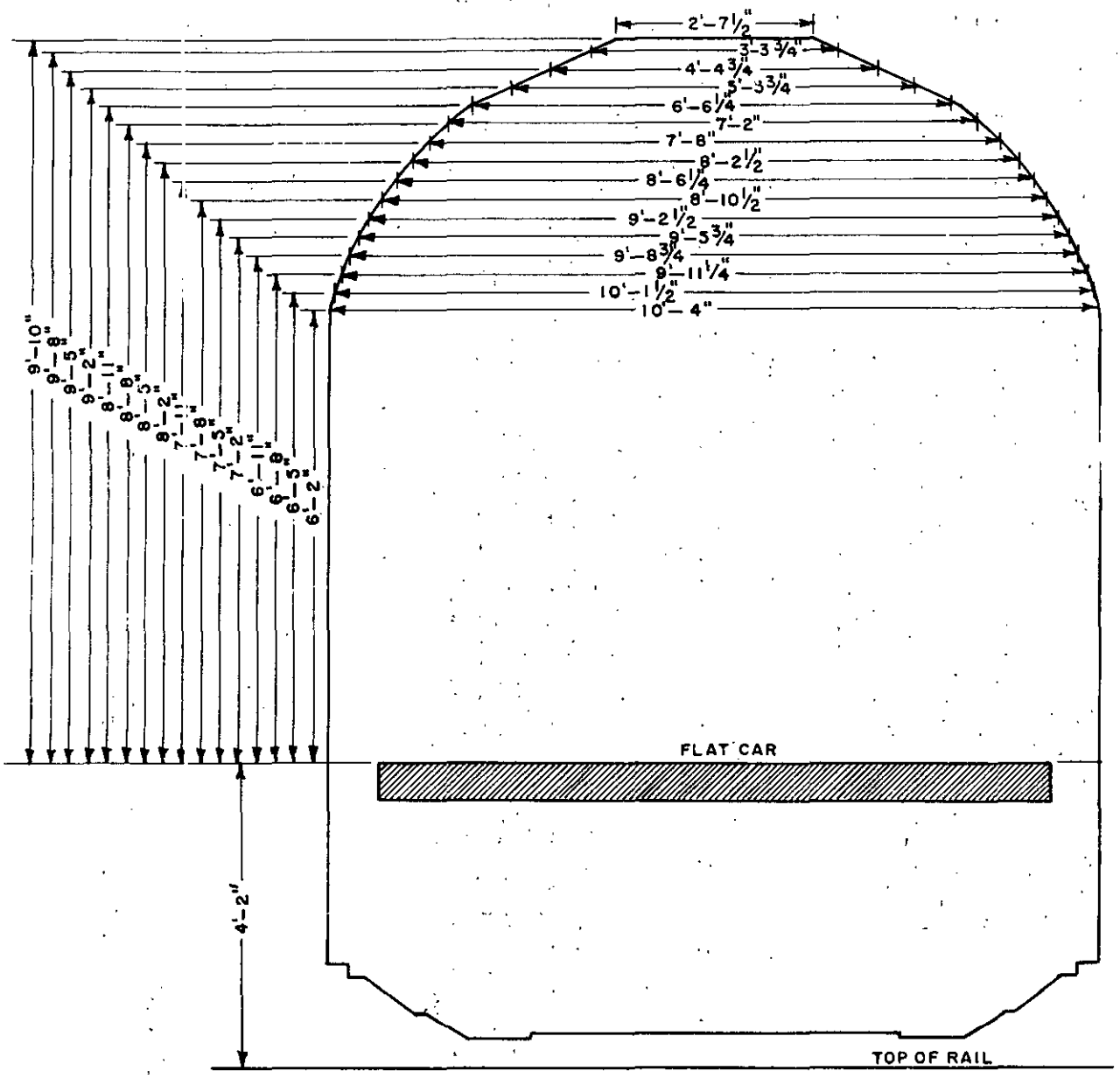


FIGURE 1. C-124 EQUIPMENT LOADING PLANNER

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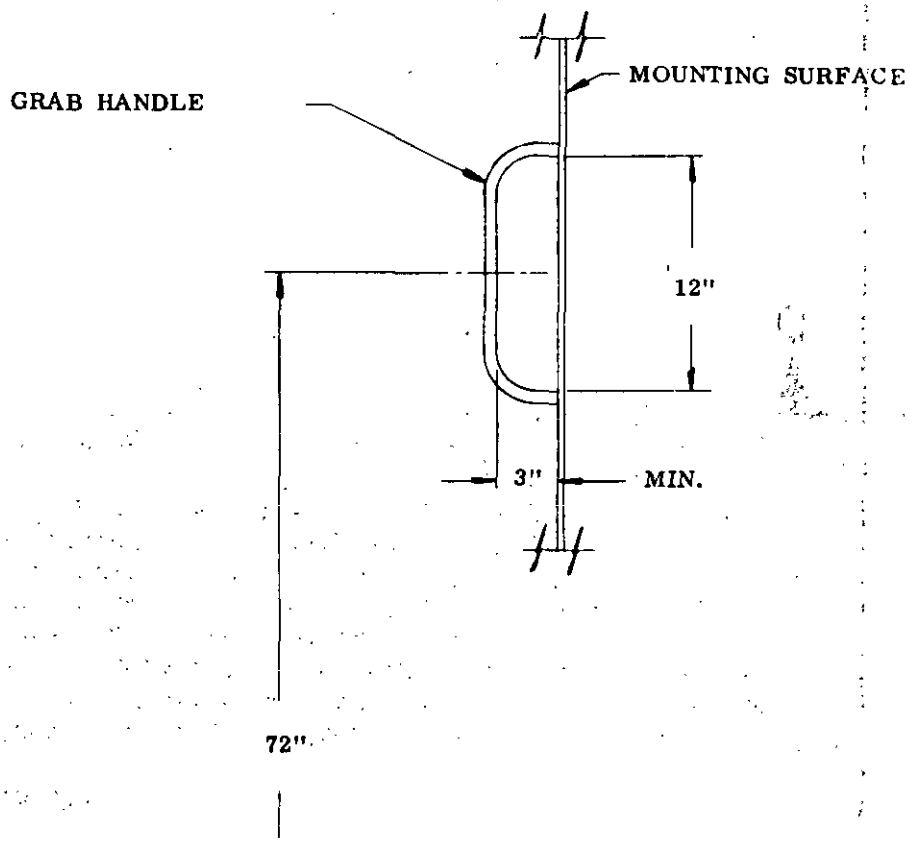


ITEM LENGTH-32'-10"

FIGURE 2. BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

A13123-1

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GRAB HANDLES

FIGURE 3.

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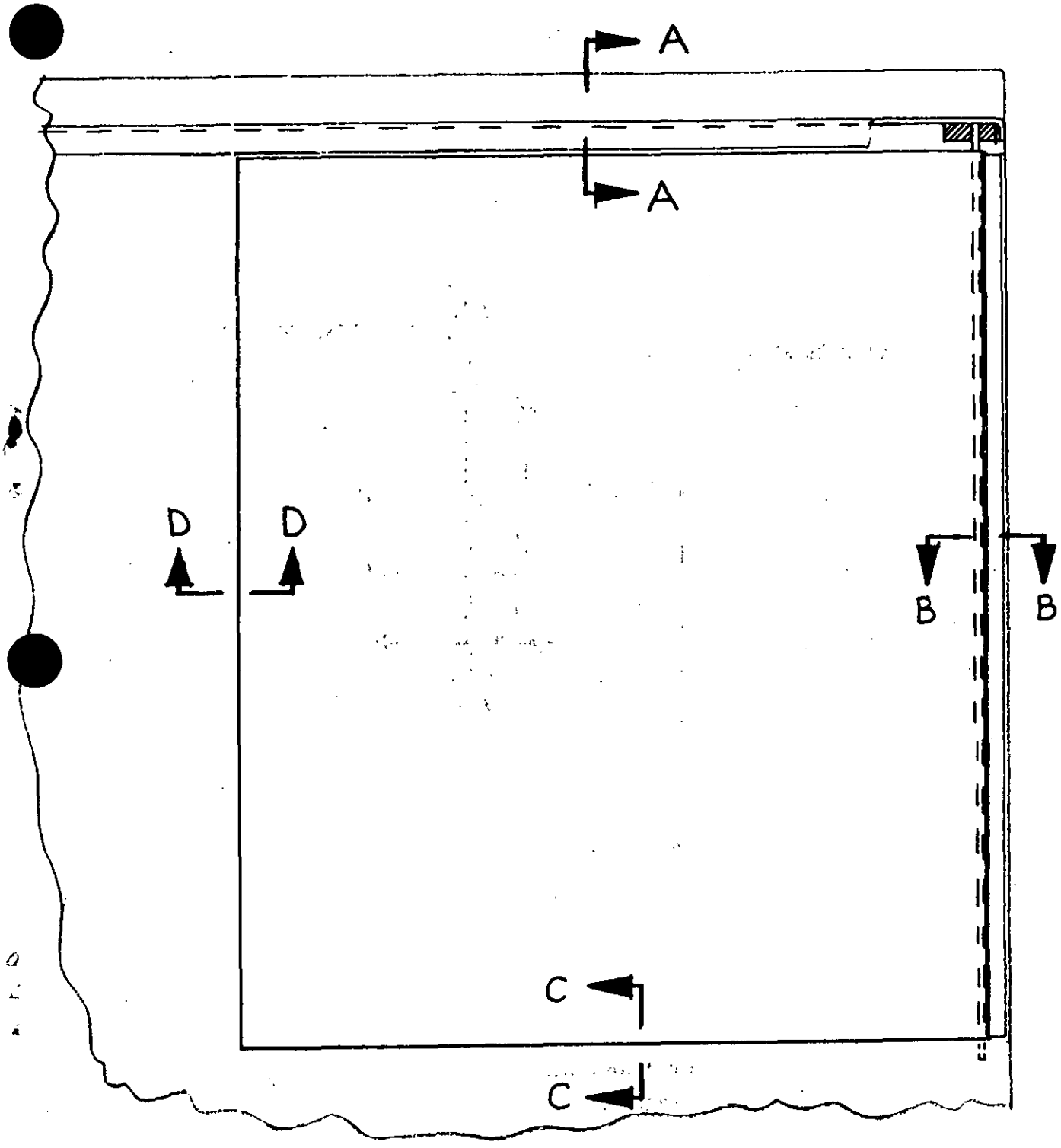


FIGURE 4. LABYRINTH SEAL TYPICAL VIEW

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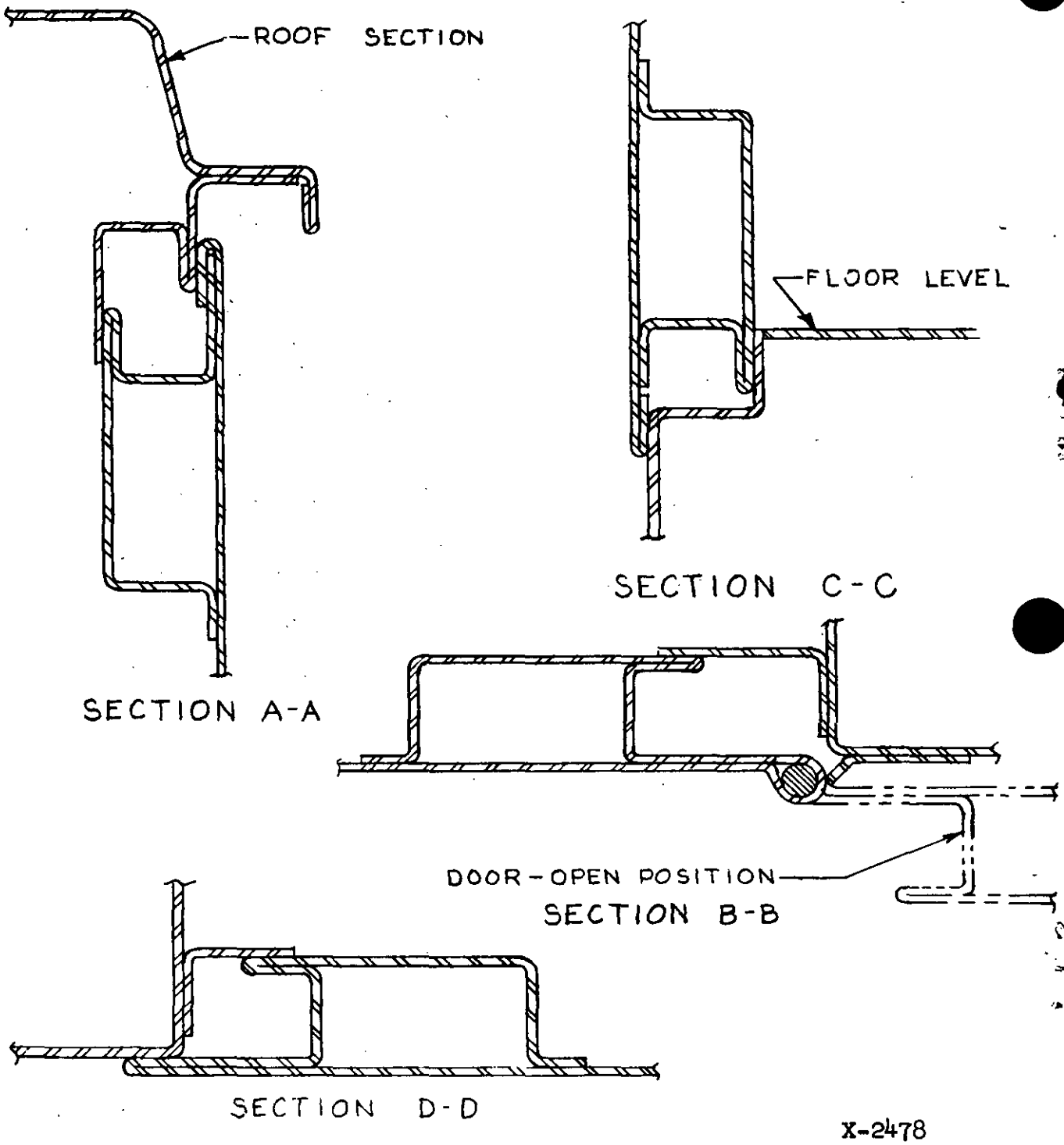


FIGURE 5. LABYRINTH SEAL TYPICAL VIEW
SECTIONAL VIEWS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-52640A(ME) - CRANE-SHOVEL, CRAWLER MOUNTED, 12-1/2-TON AT 12-FOOT RADIUS, 3/4 CUBIC YARD, DIESEL-ENGINE-DRIVEN, FOR MILITARY

ORGANIZATION OPERATIONS

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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